Improving UAV-Based LAI Estimation for Forests Over Complex Terrain by Reducing Topographic Effects on Multispectral Reflectance

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***Abstract*— Leaf area index (LAI) is a key parameter for characterizing the dynamics of terrestrial ecosystems and is also one of the important structural parameters that can be retrieved from remote sensing (RS) data. LAI over mountainous areas, however, is still difficult to retrieve reliably due to the topographical variation that introduces significant uncertainties into observed reflectance. In this article, we proposed a new scheme to estimate topographic influence on ratio-based vegetation indices (VIs) from diffuse radiation, which is not yet adequately considered in existing topographic correction schemes. In our scheme, unmanned aerial vehicle (UAV) light detecting and ranging (LiDAR) data were first used to model the sky view factor (SVF) and terrain view factor of target pixels in a slope coordinate system. Based on these view factors, the total incident solar radiation on slope (ISRS) was corrected, specifically for the diffuse sky irradiance and adjacent terrain-reflected irradiance over complex terrains. we then recalculated the multispectral reflectance of UAV images and evaluated the topographic effects on the normalized difference vegetation index (NDVI) because the magnitudes of correction on red and near-infrared (NIR) reflectances are quite different. Finally, large-scale LAI distribution was retrieved by empirical models developed from the relationships between terrain-corrected NDVI and field-measured LAI. Our results show that the proposed topographic correction scheme can significantly improve the LAI retrievals over a growing season. Given that forests are widely distributed in complex terrains around the globe, this study would have significance in improving the mapping of global LAI that is essential for terrestrial carbon cycle studies.**

***Index Terms*— Incident solar radiation, leaf area index (LAI), topographic effects, unmanned aerial vehicle (UAV).**

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NOMENCLATURE

Latin Letters

E	Total incident solar irradiance on slope [W/m^2].
E_{dir}	Direct solar irradiance [W/m^2].
E_{dif}	Diffuse sky irradiance [W/m^2].
E_{adj}	Adjacent terrain-reflected irradiance [W/m^2].
E_0	Solar irradiance at the top of the atmosphere [W/m^2].
E_s	Solar irradiance over Earth's surface [W/m^2].
E_c	Incident irradiance on adjacent terrain [W/m^2].
L_d	Sky radiance [W/m^2·sr^{-1}].
$H(\varphi)$	Maximum sky opening angles in direction φ [deg].
El	Elevation [m].
D	Plane distance [m].
A	Aspect [deg].
S	Maximum slope in the direction A [deg].
S_p	Slope in direction φ [deg].
K	Anisotropic factor of diffuse sky irradiance [−].
V_s	Sky view factor (SVF) [−].
V_c	Terrain view factor [−].
R	Reflectance [−].
f	Topographic effect [−].

Greek Letters

θ	Zenith angle [deg].
φ	Azimuth angle [deg].
τ	Atmospheric optical depth [−].

Subscripts

λ	Wavelength.
u	Unmanned aerial vehicle (UAV).
red	Red band (650 ± 16 nm).
nir	Near-infrared (NIR) band (840 ± 26 nm).
ct	Complex terrain condition.
ft	Flat terrain condition.
c	Topographic correction.

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I. Introduction

FORESTS play an important role in global carbon and water cycles as a main biome of the Earth's terrestrial biosphere [1], [2], [3], [4]. Forests are widely distributed over complex terrains, where they are usually understudied due to their complexity. For example, the topographic effect is a crucial factor that could significantly bias the characterization of key parameters and processes of forests but it is not well quantified in global ecosystem research [5], [6], [7]. As a result, the roles of forests over complex terrains in global terrestrial carbon and water cycles remain highly uncertain [8].

Leaf area index (LAI) is a crucial biophysical variable for simulating gross primary productivity (GPP) and evapotranspiration (ET) [9], [10], and its measurements can be affected by topography for many reasons related to its measuring techniques. LAI is defined as half the total green leaf area per unit horizontal ground area [11], [12]. For broadleaf vegetation, half of the total green leaf area can be simplified as one-sided green leaf area, while for conifer needles, the area of all sides is included in the total leaf area. In complex terrains, the ground area in the LAI definition can be a sloping or horizontal surface, and the sloping and horizontal surface areas have a clear projection relation [13], [14], [15]. Direct ground observation and remote sensing (RS) are common ways to obtain LAI across various spatial scales, but both are affected by topography. For ground observation, optical instruments, such as fisheye cameras, the LAI-2200 plant canopy analyzer, and tracing radiation and architecture of canopies (TRAC), are widely used to obtain indirect measurements of LAI [16]. Meanwhile, placing the instruments horizontally or parallelly to the slope will obtain different LAI values. Studies confirm that LAI values measured parallel to the slope are larger than those measured horizontally, and the difference is larger for steeper terrains (e.g., slope $>30°$) [6], [14], [17]. Because ground LAI values are usually used as "ground truth" to validate RS-based LAI retrievals, the referencing coordinate system of ground LAI measurements should be matched with that of RS data (e.g., slope coordinate system after topographic correction for complex terrains). RS-based methods usually exploit the relationship between reflectance or vegetation indices (VIs) and target LAI, and the quality of reflectance is critical to the accuracy of LAI estimation. Relative to flat terrains, the slope and aspect factors of complex terrains would additionally influence the observed reflectance and weaken the relationship between LAI and reflectance [18], [19]. Hence, reducing topographic effects on reflectance is essential to provide reliable large-scale LAI retrievals over complex terrains.

A wide range of topographic correction approaches for reflectance have been developed, which can be grouped into three broad categories: empirical, Lambertian, non-Lambertian methods, and hybrid methods. Based on the assumption that the topography changes the geometry of sun–target–sensor and contributes to variations in the incoming and outgoing irradiances [20], empirical relationships between incoming radiance and incident angle (solar illumination angle) can be built for topographic correction [21], [22]. Although empirical methods are usually simple in mathematical forms and easy to operate, they are often site- and time-specific.

Moreover, the irradiance on a horizontal surface should be known before the correction, any errors of ground observations will increase uncertainties in the corrected reflectance.

Introducing physical parameters to account for radiative transfer processes can overcome the limitation of site- or time-specific empirical methods, and a series of semiempirical or physical models have been developed. The most straightforward topographic correction is to convert the irradiance on the horizontal surface to the sloping surface based on the cosine law [23], forming the basis of a typical Lambertian method, named the cosine model [22]. However, this method often leads to overcorrection because the diffuse component of the incoming radiation is much less angle dependent. To address the overcorrection issue, some models, including the C model [22], cosine-C model [24], cosine-B model [25], and gamma model [26], were deigned to optimize the simple cosine model. The sun–canopy–sensor (SCS) model is another Lambertian method which can consider the tree geotropism [20]. Similar to the cosine-C model, an SCS + C model was also formed using a semiempirical factor C to lower the overcorrection [27]. Overall, the Lambertian methods only correct for the direct solar radiation, and the overcorrection issue cannot be completely avoided especially when the illumination angle is close to 90° [20].

Comparing with the Lambertian assumption, the assumption of non-Lambertian surfaces resembles more closely the reality. A constant factor k designed by Smith et al. [28] can be used to develop topographic correction methods suitable for non-Lambertian surfaces, e.g., the Minnaert and Minnaert + SCS model [29]. Although these models perform well for various landcover and RS data [30], their applications for large areas are still difficult since the factor k should be calibrated for different STS relations. The aforementioned topographic correction methods can reduce topographic effects for various RS data [26], [31], [32], [33]. Their performance, however, varies widely for different RS sensors, study areas, and land cover types [34], [35]. Corrections by these methods are not sufficient for generating reliable LAI products over complex terrains. An important weakness of these methods is that they commonly focus on the correction of direct solar radiation only.

Besides direct radiation, the diffuse sky irradiance and adjacent terrain-reflected irradiance can also be received in significant amounts by the target object over complex terrains [36]. To capture topographic effects on diffuse radiation, several hybrid methods have been developed by integrating atmospheric correction and topographic correction [37], [38]. Li et al. [39] proposed a physical method based on atmospheric and BRDF correction which achieved satisfactory results for orthorectification of Landsat TM images in mountain areas. Zhang et al. [40] developed a coupled reflectance retrieval algorithm by integrating dark object subtraction atmospheric correction with topographic correction. Although

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hybrid methods can correct topographic effects for diffuse radiation, their broad application is usually limited by high cost of parameter calibration.

In addition, the development of radiative transfer models provides more options for topographic corrections. Some radiative transfer models are suitable for complex terrain and achieve good performance of radiative transfer simulation over mountainous areas [23]. For example, the geometric-optical model for sloping terrains (GOST) model [41], [42] was developed based on the four-scale model [43] and can simulate diffuse radiation for an extensive slope. Huang et al. [44] proposed a clear-sky 3-D parameterization scheme and performs well for surface solar radiation simulation affected by terrain. Large-scale remote sensing data and image simulation framework over heterogeneous 3-D scenes (LESS) [45] is another typical 3-D radiative transfer model designed for the radiometric simulation of forest canopies and urban scenes. Limited by the low model running efficiency, the current application of 3-D radiative transfer models is usually to provide "truth-value" for the evaluation of corrected reflectance.

For LAI retrievals, several optical VIs have been developed based on reflectance in two or more spectral bands [46], [47], [48]. The ratio-based VIs can eliminate part of topographic effects which are assumed to be the same for different bands, e.g., most topographic effects on direct solar irradiance [49]. However, the topographic effects on the diffuse sky and adjacent terrain-reflected irradiance can be hardly eliminated by ratio-based VIs. Hence, reducing the topographic effects on diffuse sky and adjacent terrain-reflected irradiance is helpful in improving the accuracy of VI-based LAI retrievals over complex terrains. Although a series of physical models have been developed for diffuse radiation correction [50], these models usually focus on the diffuse radiation correction for a single spectral band and the wavelength-dependent topographic effect still remains unknown. Typically, the sky has high irradiance in visible and low in NIR wavelengths, while vegetated terrains reflect strongly in NIR and weakly in visible wavelengths. Correction for diffuse radiation would, therefore, make significant differences in the normalized difference vegetation index (NDVI) which depends on the ratio between NIR-to-red reflectance. It is, thus, necessary to quantify topographic effects on both direct and diffuse radiation components for terrain-corrected NDVI calculation. The topographic effect on multiband RS data or VIs caused by diffuse radiation still needs more research.

Recent studies also demonstrated that the surrounding terrain effect appears lower if a coarser spatial resolution of the digital elevation model (DEM) is used [51]. Specifically, incident solar radiation on slope (ISRS) exhibits significant differences among variable terrains with a fine-resolution DEM (e.g., $\leq$30 m), but does not show large differences with the coarser-resolution DEM (e.g., 500 m) [52]. Hence, DEM with high resolutions should be involved in correcting micro-topographical effects on ISRS, especially in recent days when the rapid development of UAV and UAV-based light detecting and ranging (LiDAR) techniques have made high-resolution DEM easily obtainable [53].

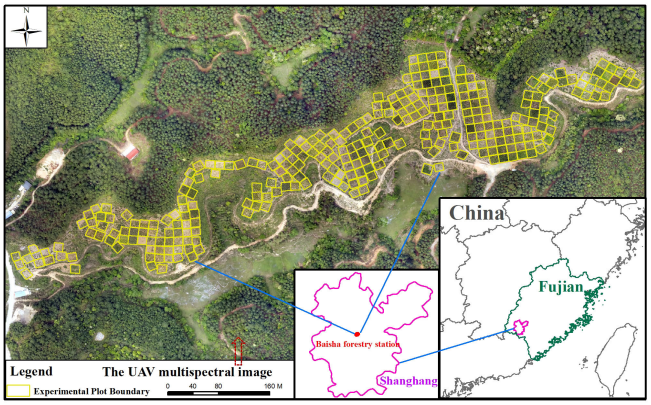


Fig. 1. Location of the study area and the distribution of experimental plots.

In this study, we attempted to correct topographic effects on ISRS for red and NIR bands, separately. The aim is to derive terrain-corrected NDVI for optimized LAI estimation in a subtropical forest over complex terrains. For this purpose, a new scheme is developed to estimate the contributions of diffuse sky and terrain reflected irradiance to the incoming shortwave radiation on sloping surfaces. As the topographic effects of red and NIR bands are different in the two irradiances, our new topographic correct scheme is capable of capturing the topographic effect on NDVI. The main objective of this study is to answer the following questions: 1) if topographic effects on ISRS differ between red and NIR bands and if the topographic correction can be embodied in the NDVI calculation? 2) if the terrain-corrected NDVI can optimize the retrievals of forest LAI over complex terrains? and 3) can the UAV help obtaining reliable time-series spectral data for forest canopies over complex terrains.

II. Materials and Methods

A. Study Area and Field Campaign

Two study areas were selected in this study: Baisha Forest Farm and Sanming Research Station of Forest Ecosystem and Global Change. The main field campaigns were conducted at an experimental site (116°42′E, 25°05′N) in the Baisha Forest Farm, Fujian, China, which is located in southeastern China (Fig. 1). The local climate is subtropical, characterized by a mean annual precipitation of 1637 mm and mean temperature of 19.8 °C between 2000 and 2020 [54]. The experimental site, which is designed for studying the relationships between forest biodiversity and ecosystem function under climate change, was slash-burned and established with 76 544 individual trees from 32 species planted in March 2019 [54]. There are 299 plots in total established with different combinations of species. Each plot is 12 × 12 m in size (grids in Fig. 1) and has 256 trees arranged in a square with 0.75-m spacing.

As shown in Fig. 2, the topography of the experimental site in Baisha Forest Farm is complicated: the maximum difference of elevation is more than 150 m (the study area is less than 1.0 × 0.3 km); the slope of this area varies from 0° to 81° with a mean of 27°. Furthermore, there are nine complete ridges and valleys in the study area. The variable microtopography is

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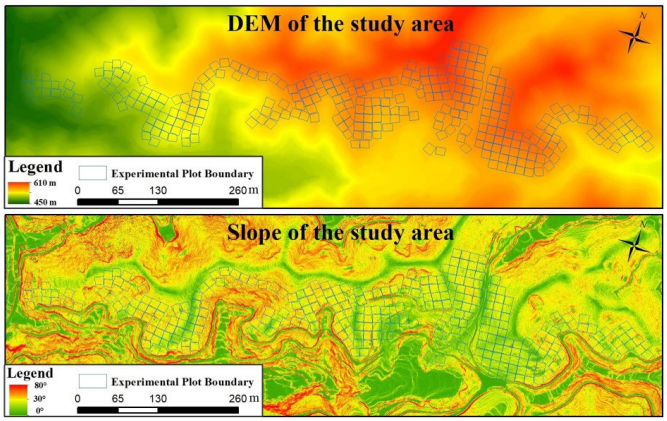


Fig. 2. Terrains of the study area in Baisha Forest Farm. (a) DEM. (b) slope.

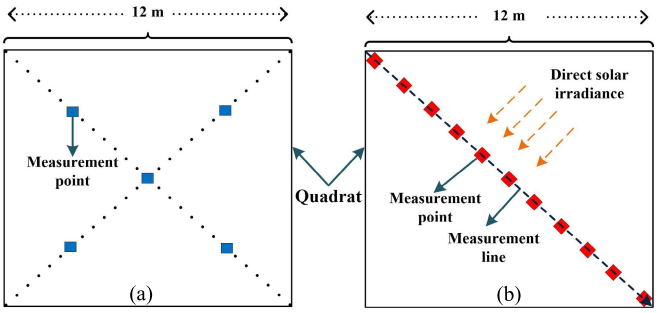


Fig. 3. Layout of quadrats and samples for (a) LAI and (b) CI.

helpful for us to analyze topographic effects on multispectral reflectance measured by UAV.

A total of five field campaigns were conducted in 2021 to collect LAI. The dates of these sequential campaigns were April 3–6, June 16–19, August 11–15, October 13–17, and December 14–18. A total of 75 plots were selected for all the five field campaigns, and an additional 45 plots were selected for the first campaign. Five observation locations in each plot were marked by polyvinyl chloride (PVC) tubes to ensure the data consistency between field campaigns. We measured effective LAI at each point with LAI-2200 Plant Canopy Analyzer (Li-Cor, Lincoln, NE, USA) and following a scheme of two above canopy (before and after below canopy measurements) and five below canopy measurements for each plot [Fig. 3(a)]. The LAI-2200 sensor was horizontally placed for both above and below canopy measurements. The time period either before 10:00 A.M. or after 4:00 P.M. was selected to minimize the effects of direct sunlight on LAI measurements. The measured LAI will be further transformed into the slope-coordinate system [LAI_s of (22)].

We also measured the clumping index (CI) using TRAC-II instrument (LICA, Beijing, China) along a transect perpendicular to the direction of solar light. The TRAC-II sensor was also held horizontally while walking and measurements were conducted between 11 A.M. and 2 P.M. to better capture the signals from direct sunlight. Similar transformation was also made to calculate CI under the slope coordinate system (CI_s). The total LAI was then calculated as effective LAI (LAI_s) divided by CI_s. The total LAI will be further transformed to

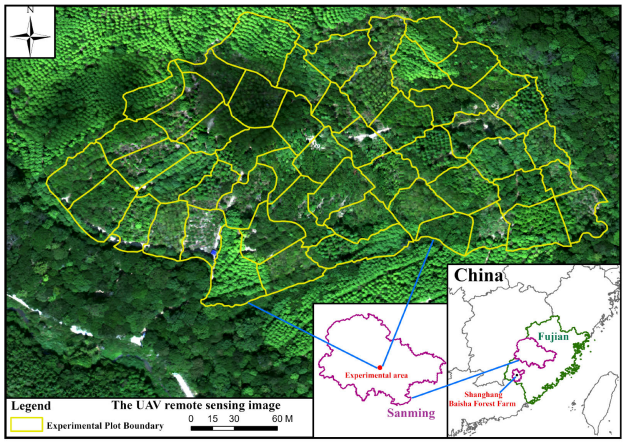


Fig. 4. Locations of the experimental site in Sanming.

the slope-coordinate system [LAI_s of (22)]. The transformed LAI was used to analyze the performance of the topographic correction scheme proposed in this study.

To evaluate the suitability of the proposed method for different areas, we selected another experimental site (117°28'E, 26°09'N) in Sanming Research Station of Forest Ecosystem and Global Change, Fujian Province, southeastern China (Fig. 4). The straight-line distance between this site and Baisha Forest Farm is more than 150 km. The experimental site is also subtropical mountainous region with a mean annual precipitation of 1610 mm and a mean temperature of 19.3 °C during 1960 and 2019 [55]. The slope of this area varies from 0° to 81° with a mean of 20°. The tree species and topographic relief of this site are both different from the study area in the Baisha Forest Farm. Similar field campaigns were conducted on September 17 (clear sky) of 2022 to collect field LAI. A total of 18 plots were selected, and their total LAI values were measured using LAI-2200C and TRAC-II.

B. RS Data Acquired by the UAV System

Two types of UAV-based RS data were acquired in this study: multispectral images for canopy reflectance and LiDAR data for DEM. The multispectral images were collected by DJI P4-Multispectral (DJ-Innovations, Shenzhen, China) that consists of a four-rotor UAV, six image sensors, an electronic component for incoming radiation, and a ground control system. The six sensors include one visible (images in JPEG format) and five multispectral sensors (images in TIFF format) with a maximum resolution of 1600 × 1300 pixels. The central wavelengths of the acquired images are 450 ± 16 nm (blue), 560 ± 16 nm (green), 650 ± 16 nm (red), 730 ± 16 nm (red edge), and 840 ± 26 nm (NIR).

In each LAI field campaign in the Baisha Forest Farm, six flights were undertaken in 1-h of interval from 10 A.M. to 3 P.M., which were frequent enough to capture both clear-sky and cloudy conditions. The flight height in this study was set to 100 m above the ground, and the overlap for the acquired image swaths was 80% both longitudinally and latitudinally. Prior to the flight, two whiteboards were placed in open areas, and their reflectance was measured by the

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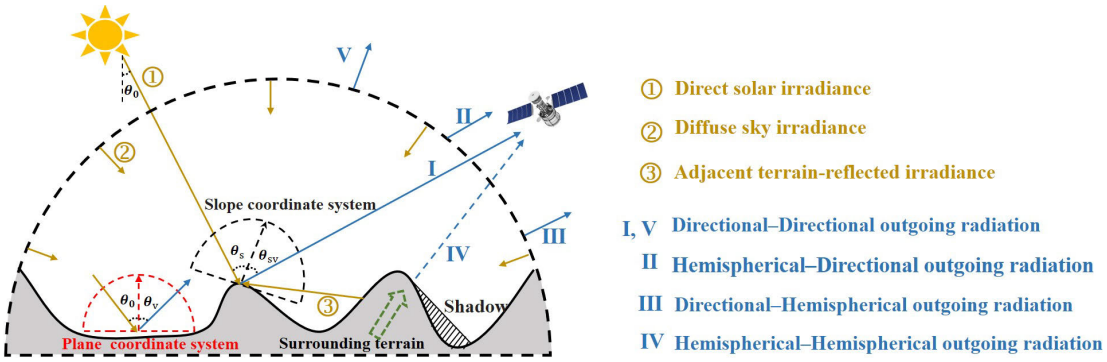


Fig. 5. Radiative transfer process over complex terrain.

Analytical Spectral Devices (ASD) FieldSpec4 for radiometric calibration. A global positioning system (GPS) with real-time kinematic (RTK) was temporally installed to receive the location information for image stitching and geometric correction.

The multispectral images were first processed for reflectance calculation, image stitching, and geometric correction by DJI Terra (https://www.dji.com/cn/dji-terra). Six band images were then obtained with a spatial resolution of 0.067 × 0.067 m. The images without stitching, which contained whiteboards, were also gathered for ISRS correction.

LiDAR data for the experimental site in the Baisha Forest Farm were collected by a LiDAR sensor (RIEGL VUX-1LR, maximum scanning frequency 820 kHz) attached to a six-rotor UAV (DJI M600). The LiDAR flight was conducted on July 22 with a flight height of 200 m and a track spacing of 50 m. After data preprocessing, the LiDAR point clouds with the density of 568 /m^2 were obtained to generate DEM after excluding the influence of plant canopies with the highest resolution of DEM is 0.25 × 0.25 m. Considering the canopy size in the study area, a spatial resolution of 0.50 × 0.50 m is suitable to reduce the effect of self-formed shadows and pure nonleaf pixels. Meanwhile, this spatial resolution is high enough for us to analyze the microtopography of the study area. Because the spatial resolution of DEM is different from the multispectral data, a bilinear-interpolation resampling method was used to change the DEM's resolution to 0.536 × 0.536 m. Then, one DEM pixel covers 8 × 8 multispectral pixels. The slope and aspect with 0.536 × 0.536 m were calculated for the study area using ArcGIS for terrain analysis.

The aforementioned UAVs were also used to acquire multispectral images and LiDAR data for the experimental site at the Sanming Research Station. The UAV flights were conducted on September 17, 2022, with the same flight parameters. After data preprocessing, the multispectral images and DEM data were obtained.

C. Algorithms for Correcting the Incident Solar Radiation Over Complex Terrain

In this section, we designed an integral process for correcting the diffuse sky and adjacent terrain-reflected irradiance correction over complex terrain using UAV data. Unlike the existing methods [36], [50], the core terrain parameters were calculated using a slope coordinate system. As shown in Fig. 5, the ISRS (E) received by the target pixel can be divided into three sources

$$E = E_{\text{dir}} + E_{\text{dif}} + E_{\text{adj}} \tag{1}$$

where E_{dir} is the direct solar irradiance, E_{dif} is the diffuse sky irradiance, and E_{adj} is the adjacent terrain-reflected irradiance. The direct solar irradiance over the flat terrain (E_s) of the study area was calculated from the slope and aspect (see Appendix A).

The diffuse sky irradiance (E_{dif}) over complex terrains can be estimated as given in the following equation [36]:

$$E_{\text{dif}} = \pi K \overline{L_d} V_s \tag{2}$$

where $\overline{L_d}$ is the mean sky radiance. Because the topographic effects cannot influence the $\overline{L_d}$ values, the multispectral UAV observations of whiteboards placed in open areas were used for $\overline{L_d}$ calibration. We first gathered the reflected irradiance of red and NIR bands. Then, the direct solar irradiance for whiteboards was calculated and removed from the UAV observations. We considered the mean value of multiangle UAV observations as the calibrated $\overline{L_d}$, which was obtained in this way for each image's topographic correction. K is the anisotropic factor of diffuse sky irradiance under the flat terrain conditions [56], [57]. In this study, we ignored the variations of K in both zenith and azimuth directions because the lack of ground observation data, and $K = 1$ was used to correct E_{dif} in red and NIR bands.

V_s is the SVF, which is defined as the ratio of sky irradiance incident on a target slope to that incident on an unobstructed horizontal surface [50]. In complex terrains, the calculations of SVF on a horizontal and a sloping plane are different. In this study, a slope-coordinate system was used to calculate the SVF of the hemispheric sky, which can be expressed as follows [58]:

$$V_s = \frac{1}{\pi} \int_0^{2\pi} \int_0^{H_{(\varphi)}} \sin\theta \cos\theta d\theta d\varphi \tag{3}$$

where $H_{(\varphi)}$ is the maximum sky opening angle relative to the normal to the sloping surface at the azimuth angle φ.

Equation (3) can be simplified as follows:

$$V_s = \frac{1}{2\pi} \int_0^{2\pi} \sin^2 H_{(\varphi)} d\varphi. \tag{4}$$

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Similar to SVF, the terrain view factor (TVF) denoted by V_c) can also be calculated using the terrain parameters

$$V_c = \frac{1}{\pi}\int_0^{2\pi}\int_{H_{(\varphi)}}^{\frac{\pi}{2}} \sin\theta\cos\theta d\theta d\varphi = 1 - V_s. \tag{5}$$

The adjacent terrain-reflected irradiance (E_{adj}) over complex terrain can be calculated as follows [59]:

$$E_{\text{adj}} = \overline{E_c}\rho V_c \tag{6}$$

where $\overline{E}_c$ is the mean incident irradiance on adjacent terrain, and $\overline{\rho}$ is the mean reflectance of adjacent terrain.

Because of the spatial variation of ground vegetation on adjacent terrain, the mean incident irradiance and reflectance might not well represent the actual spatial heterogeneity and could bring in significant errors into (6). To tackle this issue, the following equation was introduced in this study to estimate E_{adj} [50], [52]:

$$E_{\text{adj}} = \sum_{n=1}^{N}\left(L_n \cos T_m \cos T_n S_n / R_{mn}^2\right) \tag{7}$$

where N is the total count of valid pixels which can contribute the reflected irradiance to the target pixel in specific TVF, m and n are the numbering of the target and adjacent pixels, respectively, L_n is the reflected irradiance of the adjacent pixel m, S_n is the surface area of pixel n, T_m and T_n are the angles between the normals of pixels m and n and the line linking m and n pixels, and R_{mn} is the length of the straight line between pixels m and n. The values of T_m, T_n, and R_{mn} can be calculated in ArcGIS using DEM data.

D. Topographic Correction on NDVI for LAI Retrieval

Optical VIs have been widely used for LAI estimation. They have some advantages when applied to complex terrain, such as ratio-based VIs can reduce topographic effects to a large extent [49]. In this study, we used the NDVI [60] to develop empirical models for LAI estimation

$$\text{NDVI} = (R_{\text{nir}} - R_{\text{red}})/(R_{\text{nir}} + R_{\text{red}}) \tag{8}$$

where R_{red} and R_{nir} are the reflectance of red and NIR bands. The proposed ISRS correction scheme was used to generate terrain-corrected NDVI.

The UAV-based reflectance of wavelength λ in this study can be calculated as follows:

$$R_\lambda = \frac{E_{u-\lambda}}{E_{\text{ft}-\lambda}} = \frac{E_{u-\lambda}}{E_{\text{dir}-\text{ft}-\lambda} + E_{\text{dif}-\text{ft}-\lambda} + E_{\text{adj}-\text{ft}-\lambda}} \tag{9}$$

where $E_{u-\lambda}$ is the outgoing radiation which can be received by UAV sensor (Fig. 5, I and II), $E_{\text{ft}-\lambda}$ is the total ISRS under the flat terrain condition and can be obtained from the whiteboard using ASD. $E_{\text{dir}-\text{ft}-\lambda}$, $E_{\text{dif}-\text{ft}-\lambda}$, and $E_{\text{adj}-\text{ft}-\lambda}$ are the direct solar irradiance, diffuse sky, and adjacent terrain-reflected irradiance under the flat terrain condition ($E_{\text{adj}-\text{ft}-\lambda} = 0$), respectively. Thus, any variations of ISRS will be reflected in the reflectance directly. If we consider the topographic effect (f) of ISRS on each band over complex terrain, (9) becomes

$$\text{NDVI} = \frac{\left(f_{\text{nir}}R_{\text{nir}} - f_{\text{red}}R_{\text{red}}\right)}{\left(f_{\text{nir}}R_{\text{nir}} + f_{\text{red}}R_{\text{red}}\right)}. \tag{10}$$

For direct solar irradiance, the topographic effect ($f_{\text{dir},\lambda}$, see Appendix A) can be approximated as follows:

$$f_{\text{dir},\lambda} \approx \cos\theta_0 e^{-\tau/\cos\theta_0}\frac{\cos\theta_0\cos S + \sin\theta_0\sin S\cos(\varphi_0 - A)}{\cos\theta_0} \tag{11}$$

where λ is the wavelength. The formula for $f_{\text{dir},\lambda}$ calculation [(11)] indicates that $f_{\text{dir},\lambda}$ is nearly independent of wavelength, which means $f_{\text{dir,nir}} \approx f_{\text{dir,red}}$. The geotropism of vegetation will cause different orientations on the slope and flat terrain and then influence the NDVI. In this study, we ignored the geotropism and (14) can be converted back to (8) for the direct solar irradiance.

If adjacent terrains are covered by forests, a higher percentage of radiation will be reflected in the NIR band than in the red band. Hence, the ratios of NIR to red in sky irradiance would be quite different from that from adjacent terrains. NDVI depends not only on the LAI of the target pixel but also on the SVF and TVF of the target pixel. If we define the topographic effects of surrounding terrains for NIR and red bands as $f_{\text{st,NIR}}$ and $f_{\text{st,red}}$, respectively. Then, the conclusion of $f_{\text{st,NIR}} \neq f_{\text{st,red}}$ can be made for complex terrain based on the aforementioned discussions. Meanwhile, the circumsolar anisotropy part of diffuse sky irradiance can also influence the NDVI calculation. To capture this influence, we roughly calculated the ratios of NIR to red in sky irradiance with different zenith and azimuth directions for NDVI calculation.

In short, the NDVI can only eliminate topographic effects for direct solar irradiance but not for the sky diffuse and terrain-reflected irradiances. Consequently, this study focused on the topographic correction for diffuse sky and adjacent terrain-reflected irradiance.

After topographic correction, ISRS over complex terrains will be reduced due to partially blocked sky by adjacent terrains, but, meanwhile, will be supplemented by the adjacent terrain-reflected irradiance (E_{adj}). Referred to (9), the corrected reflectance at wavelength λ ($R_{\lambda-c}$) can be estimated as follows:

$$\begin{aligned} R_{\lambda-c} &= \frac{E_{u-\lambda}}{E_{c-\lambda}} = \frac{E_{\text{ft}-\lambda}\times R_\lambda}{E_{c-\lambda}} \\ &= \frac{E_{\text{ft}-\lambda}\times R_\lambda}{E_{\text{dir}-\text{ft}-\lambda} + E_{\text{dif}-c-\lambda} + E_{\text{adj}-\lambda}} \end{aligned} \tag{12}$$

where $E_{c-\lambda}$ is the corrected ISRS over complex terrain under the slope-coordinate system. $E_{\text{dif}-c-\lambda}$ is the corrected diffuse sky irradiance. Based on terrain-corrected reflectance of red and NIR bands, the NDVI was recalculated with lower topographic effects. Meanwhile, the corrected NDVI will be under the slope-coordinate system, which is the same as the observed LAI.

Detailed NDVI correction and LAI estimation processes in this study (as shown in Fig. 6) are described as follows.

Step 1: UAV data preparation. The UAV-based multispectral images and LiDAR data were acquired, and then, the LiDAR data were used to gather point cloud data for DEM. The slope and aspect information of the study area were extracted from DEM using ArcGIS.

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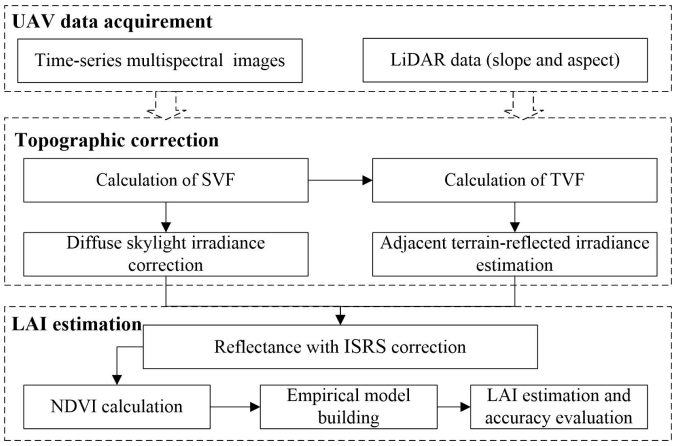


Fig. 6. Processes of LAI estimation using RS-based NDVI with ISRS corrections.

Step 2: Topographic correction for NDVI. Based on the slope and aspect derived from DEM, SVF and TVF were calculated. In this study, SVF was estimated using (44). Implementing this equation directly is difficult, thus a simplified calculation method was used as follows:

$$V_d = \frac{1}{M}\sum_{i=1}^{M}\left[\sin^2 H_{(\varphi_i)}\right] \tag{13}$$

where M is the number of azimuth angle ranges. A study has shown that a value of M between 8 and 32 is suitable for SVF calculation [61]. In this study, $M = 24$, which means that the sky hemisphere was evenly divided into 24 azimuth angle ranges (i.e., a step of 15°).

Then, (2) was used to recalculate the diffuse sky irradiance. For estimating adjacent terrain-reflected irradiance, two methods were used. In method 1, (6) was used to quantitatively estimate the irradiance from surrounding terrains without considering the spatial variation of adjacent terrain. For method 2, the term N in (7) was determined through TVF results, and then, E_{adj} was calculated using the reflectance of surrounding pixels obtained by the multispectral UAV. Based on the results of SVF and TVF, we corrected the reflectance of red and NIR bands using (12), and then, NDVI was recalculated.

Step 3: The LAI retrievals for the study area. The relationships between terrain-corrected NDVI and field LAI were developed through the empirical models. Using the empirical models and multispectral UAV data, seasonal LAI of each pixel in the study area was retrieved. Finally, the LAI retrieval accuracies using NDVI with and without ISRS corrections were compared to evaluate the performance of the topographic-effect-reducing method proposed in this study.

E. Evaluating the Accuracy of LAI Estimations

The performance of ISRS corrections for LAI estimation was evaluated using field LAI data. The mean value, coefficient of determination (R^2), and root-mean-square error (RMSE) were selected as the indices to analyze the relationship between the estimated and field-measured LAI values. The R^2 and RMSE were calculated as follows:

$$L_{\text{mean}} = \frac{1}{n}\sum_{k=1}^{n} L_{\text{obs},k} \tag{14}$$

$$R^2 = \frac{\sum_{k=1}^{n}\left(L_{\text{sim},k} - L_{\text{mean}}\right)^2}{\sum_{k=1}^{n}\left(L_{\text{obs},k} - L_{\text{mean}}\right)^2} \tag{15}$$

$$\text{RMSE} = \sqrt{\frac{\sum_{k=1}^{n}\left(L_{\text{sim},k} - L_{\text{obs},k}\right)^2}{n}} \tag{16}$$

where $L_{\text{obs},k}$ is the measured LAI values introduced above at plot k, n is the total number of the plots, L_{mean} is the mean value of L_{obs}, and L_{sim} is the estimated LAI from the aforementioned NDVI-based empirical model. R^2 and RMSE were both used to evaluate the performance of LAI estimation methods. Generally, the LAI estimation method with a higher R^2 corresponding to a lower *RMSE* was selected to be the ideal choice.

III. Results

A. Results of Adjacent Terrain-Reflected Irradiance Correction

To investigate the effect of weather conditions on ISRS corrections, we particularly selected two kinds of weather conditions to fly the drone: clear sky (around 12:30 P.M., April 6, 2021) and cloudy sky when the direct solar irradiance was shaded by a cloud but the rest sky was clear (around 10 A.M., April 4, 2021). The clear sky can enhance the terrain-reflected signals and a cloudy sky can lower the effect of direct solar irradiance. Using the proposed method (Step 2 in Section II-D), the terrain-reflected irradiance was first corrected for the UAV-based multispectral images using the mean reflectance of the surrounding terrain. Then, the reflectance of surrounding pixels obtained by the multispectral UAV was used, and terrain-reflected irradiance was recalculated. The relationship between NDVI and field LAI of April 4 [Fig. 7(a) and (b)] indicated that the terrain-reflected irradiance correction improved the LAI estimation accuracy under cloudy weather conditions, in which R^2 increased from 0.60 up to 0.65 and RMSE of field and retrieved LAI decreased from 0.45 down to 0.43. The mean value of corrected NDVI was 0.39 and lower than the uncorrected 0.48, implying that NDVI could be overestimated if topographic effects are not taken into account. Meanwhile, the consideration of the reflectance's variation of surrounding terrain is also helpful to optimize the NDVI–LAI relationship: R^2 further increased up to 0.66 [Fig. 7(c)]. Hence, the spatial variation of adjacent-terrain's reflectance was considered in Sections III-B–III-D.

Likewise, an analysis was also conducted for a clear sky condition (Fig. 8). The R^2 and RMSE values of LAI estimation without terrain-reflected irradiance correction were 0.61 and 0.45, respectively. The correction increased R^2 to 0.69 and decreased RMSE to 0.40. The mean values of uncorrected and corrected NDVI were 0.50 and 0.40, respectively. The terrain-reflected irradiance correction also improved the LAI estimation for a clear sky.

Differences also existed between the two weather conditions. The accuracy improvement by adjacent terrain correction

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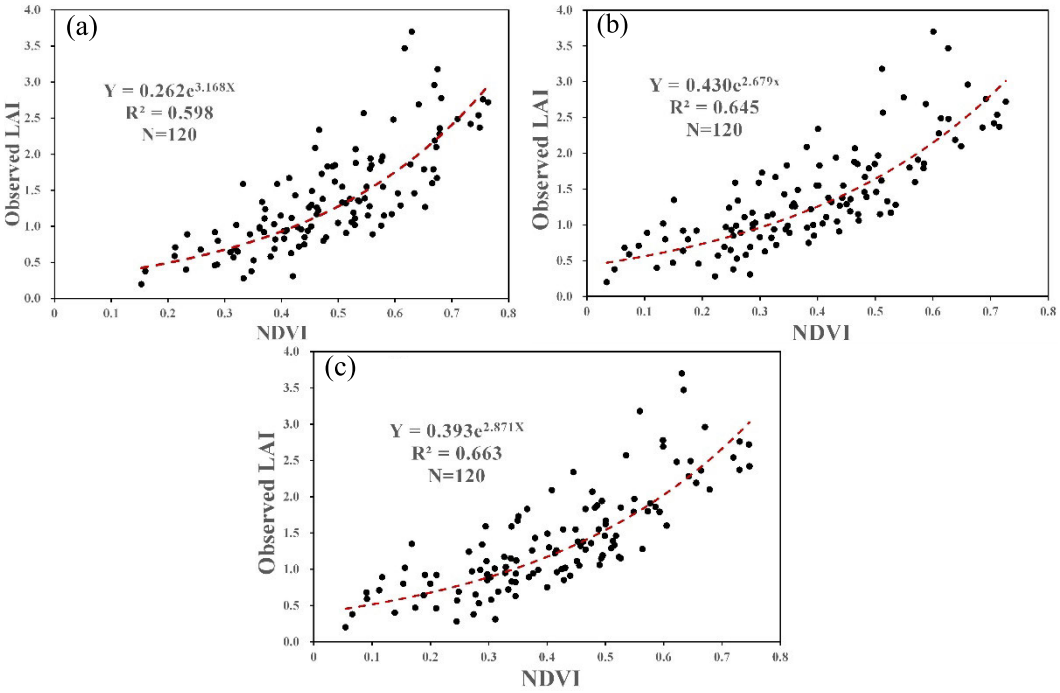


Fig. 7. Relationship between observed LAI and NDVI on April 4 under cloudy sky (a) before adjacent effect correction, (b) after adjacent effect correction using the mean surrounding reflectance, and (c) after adjacent effect correction using the reflectance of each surrounding pixel.

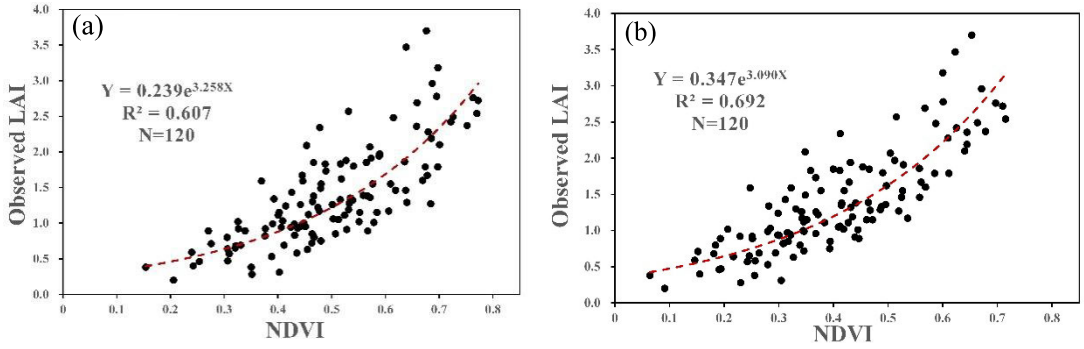


Fig. 8. Relationship between observed LAI and NDVI on April 6 under clear sky (a) before and (b) after corrections for adjacent terrain irradiance.

is higher for clear sky than for cloudy sky, and the mean NDVI values were reduced more for clear sky. For example, the decrease of mean NDVI under a cloudy sky was 0.07 after adjacent terrain correction using method 2, while the decrease under clear sky was 0.09. This is mostly because the incoming radiation of surrounding terrain under clear sky conditions is higher than the cloudy sky, and then, a relatively more terrain-reflected NIR irradiance will be received by the target pixel. After reflectance recalculation using (12), the decrease in NIR reflectance will be larger than red band.

B. Results of Diffuse Sky Irradiance Correction

Based on the adjacent terrain correction results, we then recalculated the diffuse sky irradiance for the images collected under cloudy (April 4) and clear sky (April 6) conditions. The results (Fig. 9) indicate that, as expected, the correction for diffuse sky irradiance can further improve the NDVI–LAI relationship: for cloudy sky, the diffuse sky correction increased R^2 from 0.66 up to 0.72 and decreased RMSE from 0.41 to 0.38; for clear sky, it increased R^2 from 0.70 to 0.73 and decreased RMSE from 0.41 to 0.38. The mean values of corrected NDVI were 0.37 (April 4) and 0.39 (April 6), for cloudy and clear sky, respectively, which are slightly lower than the results with adjacent terrain-reflected irradiance correction only (0.41 both for the April 4 and April 6).

Fig. 9 also indicates that the LAI estimation accuracies after ISRS corrections for cloudy and clear sky conditions were close, implying that the weather condition did not play a critical role in UAV-based LAI estimation after ISRS corrections in this study. In addition, the cloudy weather can help diffuse sky correction to achieve better optimization of LAI estimates than clear sky conditions. Compared with adjacent terrain correction, the accuracy improvement of diffuse sky correction under a cloudy sky is similar while obviously lower than that under clear sky. Because the cloudy weather of April 4 was particularly selected and not the common weather condition, the terrain-reflected irradiance correction is more important

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TABLE I
STATISTICAL MODELS FOR LAI RETRIEVALS OF BAISHA FOREST FARM

Date (month-day)	Quadrat number	Before ISRS correction	After ISRS correction
04-06	72	$LAI = 0.26 \times e^{3.15 \times \mathrm{NDVI}}$	$LAI = 0.37 \times e^{3.06 \times \mathrm{NDVI}}$
06-17	44	$LAI = 0.12 \times e^{4.03 \times \mathrm{NDVI}}$	$LAI = 0.32 \times e^{2.72 \times \mathrm{NDVI}}$
08-13	45	$LAI = 0.03 \times e^{5.57 \times \mathrm{NDVI}}$	$LAI = 0.06 \times e^{4.82 \times \mathrm{NDVI}}$
10-16	44	$LAI = 0.31 \times e^{2.95 \times \mathrm{NDVI}}$	$LAI = 0.43 \times e^{2.63 \times \mathrm{NDVI}}$
12-16	45	$LAI = 0.14 \times e^{3.87 \times \mathrm{NDVI}}$	$LAI = 0.20 \times e^{3.57 \times \mathrm{NDVI}}$

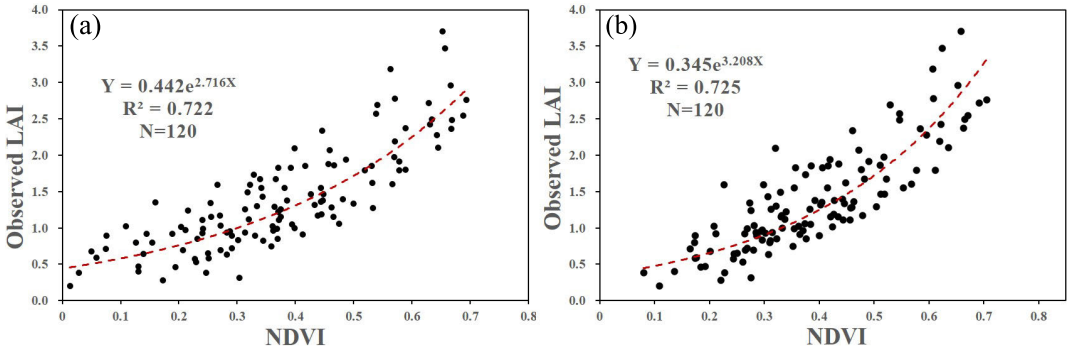


Fig. 9. Relationship between observed LAI and NDVI with both adjacent effect and diffuse sky irradiance correction. (a) NDVI for April 4 under cloudy sky. (b) NDVI for April 6 under clear sky.

for RS-based LAI estimation than the diffuse sky irradiance correction.

C. LAI Retrievals of Baisha Forest Farm

To confirm the applicability of the proposed ISRS correction scheme for seasonal conditions, we applied also it to more multispectral images collected on different dates with clear sky: June 17, August 13, October 16, and December 16. Using 60% of the observed LAI and calculated NDVI, a series of statistical models were built (listed in Table I). Then, the accuracies of these models were analyzed using UAV images and the remaining 40% of observed LAI. The evaluation results (as shown in Fig. 10) indicate that ISRS corrections can obviously optimize LAI estimations, with a 0.02–0.15 increase in R^2 and a 0.07–0.19 decrease in RMSE after ISRS corrections.

Using LAI observation in fixed 74 plots, we compared the UAV-based NDVI before and after the ISRS correction. The results [shown in Fig. 11(a)] indicate that the proposed ISRS correction algorithm can obviously lower the NDVI values. The decrease of NDVI depends on the growth stages: the mean magnitude of correction of the early- and late-growth stage (ΔNDVI = 0.10) is higher than the mid-growth stage (ΔNDVI = 0.05). Spectral saturations of the red and NIR bands are the potential reasons. These results suggest that diffuse radiation has substantial influences on NDVI. Meanwhile, the seasonal trajectories of R^2 for the LAI retrievals using the statistical models were also drawn, and the results [shown in Fig. 11(b)] indicate that the accuracies of LAI retrieval on mid-growing season dates are lower than those on other dates, also implying an effect of spectral saturation on LAI retrieval.

Meanwhile, the radiation modeling over complex terrains [25] shows that the proportion of diffuse sky and adjacent-terrain irradiance varies among different seasons. For example, the adjacent-terrain irradiance in visible bands would be higher in spring and winter with senescent vegetation or snow cover than in other seasons with green vegetation, However, this variation is not embodied in the accuracy analysis results of this study (as shown in Fig. 11). There are two potential reasons for this case: the subtropical climate and the seasonal variations of vegetation growth. The study area is in subtropical zone, and evergreen trees are the dominant tree types. The seasonal variation of adjacent-terrain reflectance at our study site would, therefore, be small relative to deciduous forests. In subtropical forests, LAI is higher in summer than in the other seasons, meaning that more NIR and less red irradiance will be reflected by adjacent terrains in the summer. To lower the influence of LAI variation, we replaced the LAI of adjacent pixels with the annual mean value and recalculated the reflectance of red and NIR bands for LAI estimation. Then, the accuracy improvements of LAI retrievals with seasonal LAI variation were assessed. The results indicate that the accuracy improvement of LAI retrievals for August 13 (R^2 increment = 0.07 and RMSE decrement = 0.09) is lower than those for June 17 (R^2 increment = 0.13 and RMSE decrement = 0.21) and October 16 (R^2 increment = 0.08 and

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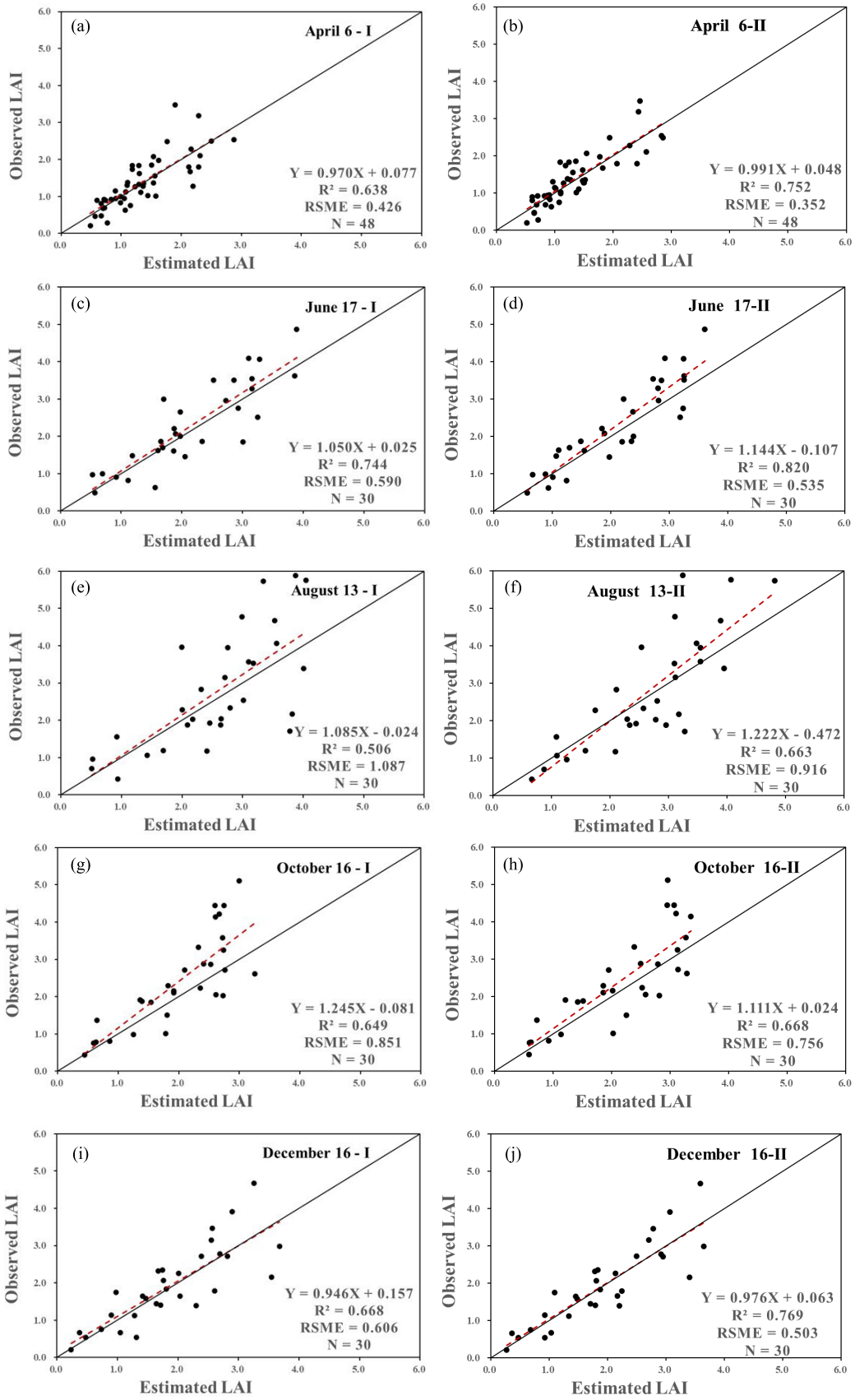


Fig. 10. LAI estimation accuracies (a), (c), (e), (g), and (i) before and (b), (d), (f), (h), and (j) after corrections using the ISRSs algorithm.

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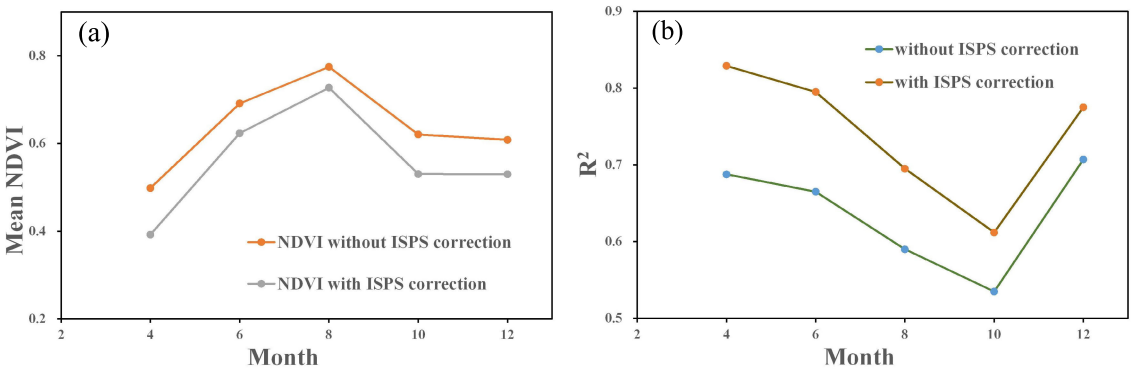


Fig. 11. Seasonal variations of mean (a) NDVI and (b) LAI estimation accuracies in the study area of 2021.

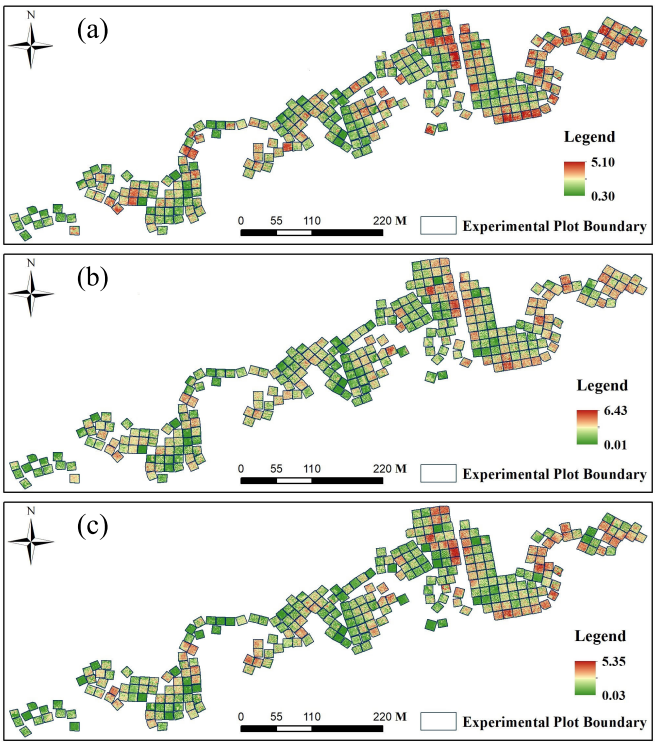


Fig. 12. LAI mapping over experimental plots in the study area (a) April 6, (b) August 13, and (c) December 16.

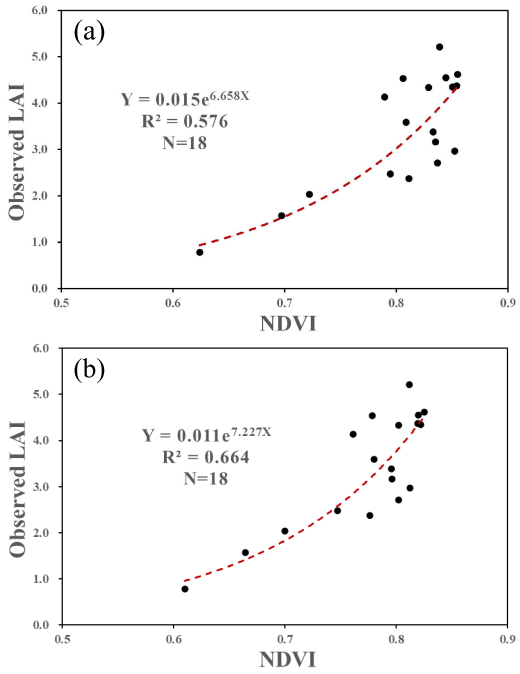


Fig. 13. Relationships between observed LAI and NDVI on September 17 (a) before and (b) after corrections using the ISRSs scheme.

RMSE decrement = 0.08). These results are consistent with our expectation that the accuracy improvement of the middle growth season should be lower than the other stages after considering the seasonal variations of LAI.

Using the NDVI–LAI relationships established with ISRS corrections, we mapped the LAI for all the 299 plots of the study area in the Baisha Forest Farm using UAV images. The results (Fig. 12) suggest that the spatial variation exists among different tree species. In addition, the results also indicate the seasonal leaf growth and senescence for the typical tree species in the subtropical mountainous areas.

D. LAI Retrievals at the Sanming Research Station

To analyze the performance of the proposed topographic correction scheme for different areas, the diffuse sky and adjacent terrain-reflected irradiance were also corrected for the multispectral images of the experimental site at the Sanming Research Station. Then, the relationships between NDVI and field LAI before and after topographic correction were built, and the results (as shown in Fig. 13) indicate that the NDVI–LAI relationship was optimized by the proposed topographic correction scheme: R^2 increased from 0.58 up to 0.68 and RMSE decreased from 0.77 to 0.68. The results confirmed that the proposed ISRS correction scheme can perform well at both two experimental sites. The high stability of the proposed scheme is important for large-scale LAI retrievals, which are useful to address issues related to terrestrial carbon and water cycles affecting global climate.

IV. Discussion

A. Advantages of the Proposed ISRS Correction Scheme

In this study, the ISRS was corrected for LAI retrieval by reducing topographic effects. The use of UAV multispectral and LiDAR data improved the spatial resolution of LAI

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TABLE II
PERFORMANCE OF SCS + C MODEL AND THE PROPOSED ISRS SCHEME FOR LAI RETRIEVALS

Method	Mean Red*	Mean NIR*	Mean NDVI	R^2	RMSE
Original data	0.21	0.55	0.50	0.61	0.45
After SCS+C model	0.20	0.53	0.50	0.62	0.45
After the proposed ISRS scheme	0.18	0.38	0.39	0.73	0.38

*Mean Red and NIR are the mean reflectance of red and NIR bands of the experimental sit in Baisha Forest Farm, respectively.

retrievals and topographic parameters. For example, the DEM data derived from LiDAR data supported us to analyze and correct inner microtopography effects for small 12.5 × 12.5 m plots.

The topographic correction for ratio-based VIs is the main advantage of the proposed ISRS correction scheme. A series of empirical or physical models have been developed for topographic correction [20], [21], [29], [50]. These models can be roughly grouped into two types: considering the diffuse radiation or not. Because the ratio-based VIs can eliminate most of the topographic effects on direct solar radiation, diffuse radiation correction will be necessary for ratio-based VIs optimization. Using the SVF and TVF derived from DEM, the topographic effects on the diffuse sky and adjacent terrain-reflected irradiance of single spectral band have been quantified by several algorithms [39], [40], [62]. However, the topographic correction for ratio-based VIs was still not well evaluated in the existing studies.

In this study, we comprehensively analyzed the topographic effects on NDVI and developed a new scheme to capture the topographic correction on NDVI. Based on the multispectral UAV images and field LAI, we compared the SCS + C model [27], which is designed based on the radiation geometrical projection relationship and the new topographic correction scheme. The results (listed in Table II) indicate that: 1) the topographic correction of the SCS + C model made no impact on NDVI although the reflectance of red and NIR bands was changed; 2) the proposed ISRS scheme can lower the topographic effects on both the single spectral band and NDVI; and 3) the accuracy of LAI retrieval after topographic correction using the proposed ISRS scheme is obviously higher than those using the original uncorrected data and the SCS + C model. The accuracy evaluation results confirmed the advantages of the proposed scheme for topographic correction of NDVI and LAI retrieval.

B. Limitations of the Proposed ISRS Correction Method

Although ISRS corrections can help providing better LAI estimates of subtropical forests in the mountainous area, there are still areas for further improvement. In this section, we summarized three uncertainties of the proposed method which are worth more research in the future.

1) Uncertainties in Field Observations for ISRS Correction and LAI Retrieval: Ground observations are the direct source of data for the algorithm development and validation. Therefore, the representation and integrity of the field observation are critical for the subsequent studies. Compared with horizontal surfaces, the high spatial heterogeneity of complex terrains will make the observations complicated. How to arrange the related instruments over complex topography remains an open question [63], [64]. In this study, two kinds of instruments were used: whiteboards, ASD, and multispectral UAV for ISRS corrections; LAI-2200 and TRAC for LAI.

In this study, we ignored the spatial variations of diffuse sky irradiance E_{dif}. However, affected by atmospheric scattering and surrounding terrain, E_{dif} should be anisotropic [56]. Hence, the use of $K = 1$ could bring in errors into E_{dif} correction. The calculation of K in direction (θ, φ) can be expressed as follows [57]:

$$K(\theta, \varphi) = \frac{L_d(\theta, \varphi)}{L_d} \tag{17}$$

where L_d (θ, φ) is the diffuse sky radiance in direction (θ, φ).

For K (θ, φ) calibration, L_d (θ, φ) is necessary, which cannot be well measured using UAV or ASD with a Lambert reflector. Besides a standard whiteboard, we also designed a whiteboard with approximate specular reflection for K calibration. Using a specific route with higher overlap both longitudinally and latitudinally and the control of the cloud platform, the reflected irradiance of the specular whiteboard with different view angles was obtained. In this study, we gathered the reflected radiance with eight azimuth angle ranges (i.e., a step of 45°) and then interpolated to 24 azimuth angle ranges ($M = 24$). Using these data, $K(\varphi)$ can be approximately calibrated for SVF calculation, and TVF was calculated using (5) from SVF. The maximum valid ranges of calibrated $K(\varphi)$ values for the whole growth season in this study are 0.31–2.01 (red band) and 0.22–2.43 (NIR band). Using the specular board, the reflected radiance with eight azimuth angles can be observed. Then, the calibration accuracy of $K(\varphi)$ was analyzed, which shows that the UAV-observations are sufficient to calibrate K values without involving the solar zenith angle ($R^2 > 0.85$ and RMSE < 0.15). Using the well-calibrated $K(\varphi)$ [even $K(\theta, \varphi)$] for ISPS correction is helpful in improving LAI retrievals. However, limited by the material and technology, the quality of the specular panel is lower than standard panel, and errors cannot be avoided completely. Multiangle measurements for diffuse sky radiance with new instruments and methods deserve a further study.

For UAV observations, we also placed two whiteboards in the open area. By taking advantage of the UAV's multiangle observation capability (high overlap of multispectral images), the distribution of diffuse sky irradiance was measured. However, the measured data can only support the calibration of its spatial and temporal variations with the azimuth angle $[K(\varphi)]$ but not with the zenith angle $[K(\theta)]$, which might have influenced the diffuse sky irradiance correction to a

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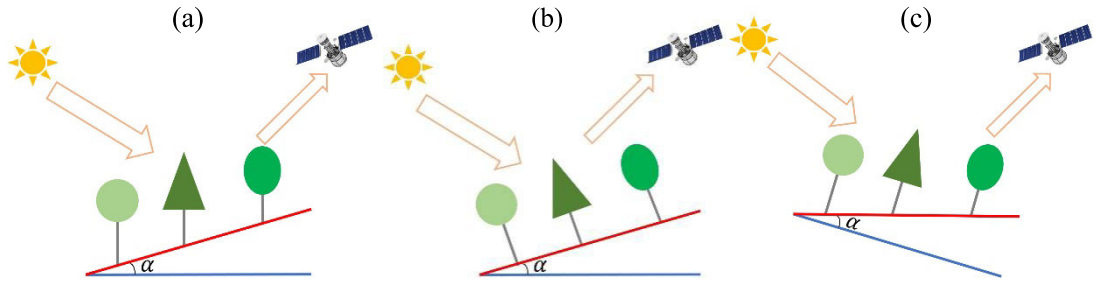


Fig. 14. Projection of LAI measured parallel to the slope to the vertical direction measured horizontally. (a) Trees vertical on the slope. (b) Trees vertical to the slope. (c) Slope with non-vertical trees.

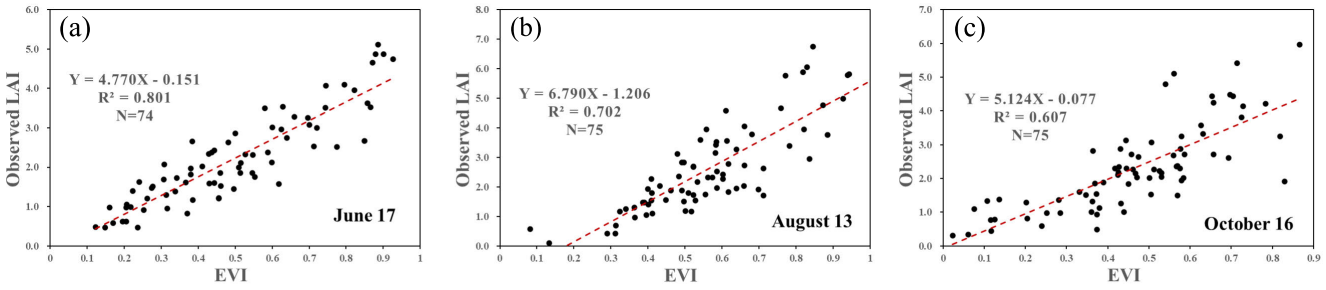


Fig. 15. LAI estimation models using EVI after ISRS corrections (a) June 17, (b) August 13, and (c) October 16.

small extent. Some more multispectral UAV observations for whiteboards with varied flight heights and more overlaps are necessary to calibrate $K(\theta)$. In addition, more whiteboards laying on the slope will be helpful to further evaluate the corrected reflectance.

The combined use of LAI-2200 and TRAC is commonly accepted for total LAI measurement [65]. In this study, we horizontally placed the sensor head of LAI-2200 during the diffuse light measurement. There are two advantages for this mode of operation: easier to control the attitude of LAI-2200 and to match RS-based LAI in the nadir direction without extra projection transformation. However, the diffuse light measurement in this way is influenced by the topography, when the slope is larger than 15°. Placing the sensor parallel to the slope can lower topographic effects, which deserves an attempt in the future study. As shown in Fig. 14, the definition of ground LAI for this method will be different from the RS-based LAI. A simple projection transformation can be made to unify the two LAI as follows [13], [15]:

$$\mathrm{LAI}_v = \mathrm{LAI}_s / \cos\alpha \tag{18}$$

where α is the slope, LAI_v is the LAI projected in the vertical direction, which can be RS-based LAI without topographic correction or ground LAI measured with a sensor placed horizontally, and LAI_s is the LAI projected to the slope, which can be RS-based LAI after topographic correction or measured LAI using a sensor parallel to the slope.

LAI measured parallel to the slope ignores the orientation of the trees: the measurement assumes that trees are vertical to the slope [Fig. 14(b)], while the fact is that trees should be vertical on the slope [Fig. 14(a)]. After projection transformation, the trees will become nonvertical [Fig. 14(c)]. Some more special LAI observations, for example, LAI measurements using both methods for the same sample plots, will be useful for us to further compare the two measurement methods and find the ideal way to gather LAI information on sloping surfaces.

2) Saturation of NDVI in Middle Growth Season: Accurate seasonal LAI information is necessary to simulate carbon and water fluxes of forest ecosystems [1], [2], [3]. As shown in Fig. 11, the ISRS correction scheme proposed in this study can improve the retrieval of an LAI time series. However, Fig. 11 also indicates that the accuracies of August and October (middle growth season) are obviously lower than those in other months, possibly due to the saturation of NDVI at high LAI values. And, the exponential relationship between NDVI and ground LAI (as shown in Fig. 10) suggests NDVI saturates at high-LAI values. To tackle this issue, a two-band enhanced VI (EVI2) has been designed [41], which can be expressed as follows:

$$\mathrm{EVI2} = \frac{2.5 \times (R_{\mathrm{nir}} - R_{\mathrm{red}})}{R_{\mathrm{nir}} + 2.4 \times R_{\mathrm{red}} + 1} \tag{19}$$

where R_{red} and R_{nir} are the reflectance of red and NIR bands. Using the multispectral UAV data after ISRS corrections, the RS-based LAI was estimated, and the accuracy analysis results for June 17, August 13, and October 16 are shown in Fig. 15. Although the saturation issue can be greatly alleviated, there are not obvious improvements for the LAI estimations. Based on (6), the topographic effect (f) for EVI2 over complex terrain will be

$$\mathrm{EVI2} = \frac{2.5 \times \left(f_{\mathrm{nir}} R_{\mathrm{nir}} - f_{\mathrm{red}} R_{\mathrm{red}}\right)}{f_{\mathrm{nir}} R_{\mathrm{nir}} + 2.4 \times f_{\mathrm{red}} R_{\mathrm{red}} + 1}. \tag{20}$$

Here, topographic effects cannot be reduced even when $f_{\mathrm{st,nir}} = f_{\mathrm{st,red}}$, meaning that the EVI2 results of (19) and (20) will be different after the topographic correction proposed in this article. In other words, the ability to reduce topographic effects for EVI2 is lower than NDVI because NDVI is strictly based on the ratio between NIR and red reflectance, while

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TABLE III
ACCURACIES FOR LAI RETRIEVALS USING DIFFERENT SVF CALCULATION METHODS

Date	Proposed method		M method		D method		L-G method	
Month-day	R^2	RMSE	R^2	RMSE	R^2	RMSE	R^2	RMSE
04-06	0.75	**0.35**	0.72	0.37	0.74	0.36	**0.76**	0.36
06-17	**0.82**	0.54	0.79	0.54	0.82	**0.51**	0.82	0.51
08-13	0.66	0.92	0.64	0.95	0.65	0.91	**0.67**	**0.90**
10-16	0.67	0.76	0.64	0.79	0.68	**0.75**	**0.68**	0.76
12-16	**0.77**	**0.50**	0.73	0.53	0.76	0.50	0.77	0.51

EVI2 is not. Taking advantages of hyperspectral images [66], it is possible to develop new VIs to reduce the saturation effect while minimizing topographic effects like NDVI. Meanwhile, LiDAR also has the potential to overcome VI saturation for regional LAI retrieval [67], [68], [69].

C. Reference Coordinate System for SVF Calculation

As shown in Section II-C, the core process for ISRS correction is the SVF calculation. In this study, the SVF was calculated in the slope coordinate system directly. Besides this system, the plane coordinate system can also be used [70]. As shown in Fig. 16, the calculations for SVF under plane and slope coordinate systems are different. Here, we selected the other three methods to evaluate the SVF calculations with different assumptions: 1) the Dozier (D) method [36], which estimates SVF using a plane coordinate system, as shown in (C1)–(C3); 2) the Manners (M) method [71], which defines SVF in a slope coordinate system, as shown in(C4) and (C5); and 3) the Lindberg–Grimmond (L–G) method [72], which is based on a shadow-casting algorithm, as shown in (C6).

Based on the DEM generated from UAV–LiDAR data, we also applied these three methods to the study area in Shanghang, and pixel-scale SVFs were estimated. Then, the SVFs were used for ISRS correction using the proposed new scheme. Finally, the accuracies of time-series LAI estimates were analyzed, and the results (listed in Table III) indicate that all these methods can well capture the variations of SVF over complex terrain. Although these methods were developed for specific conditions, the differences of R^2, and RMSE among them in this study are less than 0.03 and 0.05, respectively.

In addition, the SVF calculation method proposed by this study deserves further attention because it may be regarded as a general algorithm for a wide range of applications. The proposed method can estimate the sky opening angle range under the slope coordinate system directly. For the plane coordinate system, angle rotation cannot be avoided, complicating the calculation process. Defining SVF in the slope coordinate system can simplify parameter extraction. However, the maximum sky opening angle $H_{(\varphi)}$ still needs to be calculated in the plane coordinate system and then transformed into the slope coordinate system in theory [see (B4)].

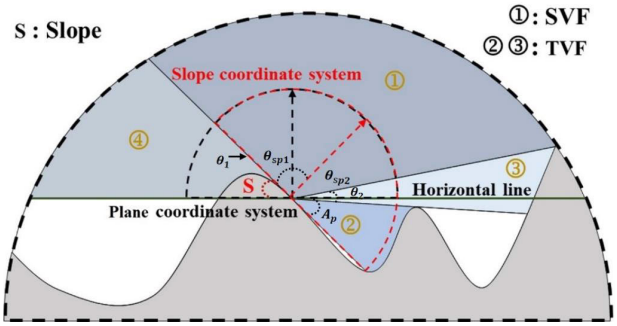


Fig. 16. Differences between slope and plane coordinate systems for SVF and TVF calculations in a specific azimuth angle φ.

As shown in Table III, the proposed method performed better than the M method for the whole growth season, and avoided errors from rotating the coordinate. We would expect larger differences for more variable terrains with steeper slopes, and the proposed physics-based method would be applicable to all sloping surfaces.

Large-scale LAI retrieval is necessary for ecological modeling. Although this study focused on reducing topographic effects on LAI retrieval over a small area by taking advantage of UAV technology, the proposed approach would be relevant to global LAI estimation. First, the study area in the Baisha Forest Farm belongs to the subtropical mountainous region, and the accuracy of LAI estimate over complex terrain is obviously lower than that over flat terrain because topographic effects have not yet been considered in the generation of many LAI products [13], [73]. The ISRS correction scheme proposed in this study can be used to improve LAI production over complex terrain, especially at high resolutions. Moreover, the use of common VI (NDVI) means that the proposed scheme is suitable for satellite RS data (e.g., Sentinel, Landsat, or GF series), which can provide large-scale multispectral data. Finally, accurate measurements of terrain elevation are a prerequisite for making the diffuse sky and adjacent terrain corrections on reflectance observed over complex terrain. With the development of spaceborne LiDAR, DEM with high spatial resolutions is available globally [74], making it possible to apply the ISRS correction methodology for large regions.

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In short, the proposed scheme can play an important role in LAI retrieval for global terrestrial ecosystem research.

V. Conclusion

This article proposed a new topographic correction scheme for RS-based LAI estimation over complex terrains. In this scheme, terrain parameters were derived using UAV LiDAR data. Then, the canopy reflectance acquired over a complex terrain using a multispectral UAV sensor was corrected for different topographic effects on incoming diffuse radiation from the sky and reflected radiation from the adjacent terrain. Because the irradiance from the sky and the adjacent terrains have different ratios in red and NIR bands, NDVI calculated based on these two bands can be influenced by the topographic correction. Using monthly ground LAI observations, empirical relationships between LAI and topographically corrected NDVI were built for seasonal LAI estimation. Through comparing the LAI retrieval accuracies with and without the topographic correction on incident solar radiation on slopes (ISRSs), we draw the following conclusions.

1) Both corrections for the diffuse sky and adjacent terrain-reflected irradiance can improve the LAI retrieval.
2) The accuracy improvement achieved by adjacent terrain correction is higher than that of diffuse sky correction, and adjacent terrain correction is more effective under clear sky than cloudy sky conditions.
3) The multispectral UAV can provide reliable time-series LAI with seasonal variations, although the accuracy of the mid-growing season is lower than those at other periods of the season because RS signals tend to be saturated at high LAI.

Appendix A
Calculation of Direct Solar Irradiance on a Sloping Surface

The direct solar irradiance over flat terrain E_s can be calculated using the following equation [36]:

$$E_s = E_0 \cos\theta_0 e^{-\tau/\cos\theta_0} \tag{A1}$$

where θ_0 is the solar zenith angle calculated in the plane coordinate system, E_0 is the solar irradiance at the top of the atmosphere, which depends on the solar constant and zenith angle to the surface, and τ is the atmospheric optical depth. On a sloping surface, E_{dir} will be

$$E_{\text{dir}} = \Delta E_s \frac{\cos\theta_s}{\cos\theta_0} \tag{A2}$$

where Δ is a two-valued function, with 0 for pixels in topographic shadow and 1 otherwise; θ_s is the solar zenith angle relative to the normal to the sloping surface. Here, we define an ISRS(E) with the zenith and azimuth angle of (θ_0, φ_0) in the plane coordinate system, the density distribution for E will be

$$\begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = E \times \begin{bmatrix} \sin\theta_0\cos\varphi_0 \\ \sin\theta_0\sin\varphi_0 \\ \cos\theta_0 \end{bmatrix} \tag{A3}$$

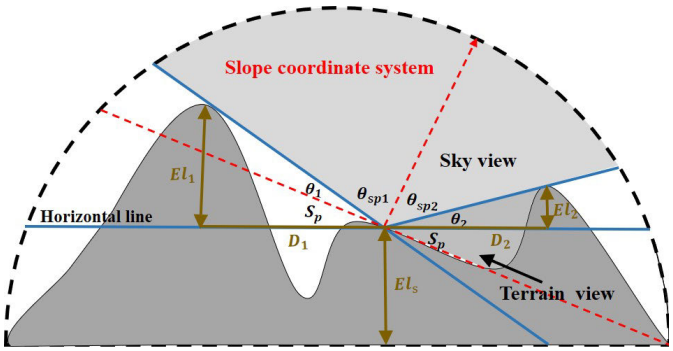


Fig. 17. Slope coordinate system for SVF calculation. θ_{sp1} and θ_{sp2} are the maximum sky opening angles [$H(\varphi)$] in the up- and down-slope directions at azimuth angle φ, respectively. S_p is the slope in direction φ, El_1, and El_2 are the elevation differences of surrounding terrain and target, D_1 and D_2 are the plane distance between the target pixel and surrounding mountain peaks, and El_s is the elevation of the target pixel.

where (x, y, z) are the coordinate axes of the plane coordinate system. For complex terrain, the slope coordinate system (S) can be transformed from the plane coordinate system (P) using the following equations:

$$S = T \times P \tag{A4}$$

$$S = \begin{bmatrix} \sin\theta_s\cos\varphi_s \\ \sin\theta_s\sin\varphi_s \\ \cos\theta_s \end{bmatrix}, \quad T = \begin{bmatrix} \cos S & 0 & -\sin S \\ 0 & 1 & 0 \\ \sin S & 0 & \cos S \end{bmatrix}$$

$$P = \begin{bmatrix} \sin\theta_0\cos(\varphi_0 - A) \\ \sin\theta_0\sin(\varphi_0 - A) \\ \cos\theta_0 \end{bmatrix} \tag{A5}$$

where θ_s and φ_s are the zenith and azimuth angles, respectively, in the slope coordinate system, and A and S are the slope and aspect, respectively. Then, θ_s can be calculated as follows:

$$\cos\theta_s = \cos\theta_0\cos S + \sin\theta_0\sin S\cos(\varphi_0 - A) \tag{A6}$$

where φ_0 is the solar azimuth angle, and A and S are the slope and aspect, respectively.

Appendix B
Calculation of Maximum Sky Opening Angles

As shown in Fig. 17, the maximum sky opening angles [$H_{(\varphi)}$] in any directions were calculated using a slope coordinate system in this study.

The calculations of θ_{sp1} in the upslope direction in the slope coordinate system are as follows:

$$\tan(\theta_1 + S_p) = \frac{\text{El}_1}{D_1} \tag{B.1}$$

$$\theta_{\text{sp1}} = \frac{\pi}{2} - \theta_1 = \frac{\pi}{2} - \arctan\left(\frac{\text{El}_1}{D_1}\right) + S_p \tag{B.2}$$

where H_1 and D_1 can be obtained from the DEM image.

The definition of S_p is the slope in any azimuth angle. As shown in Fig. 18 [58]

$$\tan S = \frac{AB}{OB} \tag{B.3}$$

$$\tan S_p = \frac{CD}{OD} \tag{B.4}$$

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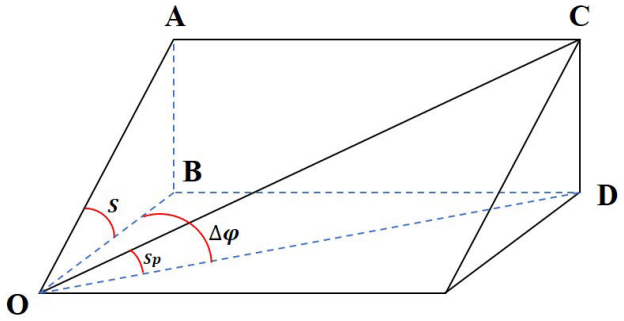


Fig. 18. Relationship between the maximum slope S in aspect A and slope S_p in any azimuth angle φ.

$$\mathrm{CD} = AB \tag{B.5}$$

$$\mathrm{OD} = \frac{\mathrm{OB}}{\cos(\Delta\varphi)} = \frac{\mathrm{OB}}{\cos(\varphi - A)} \tag{B.6}$$

where A is the aspect, S is the slope in the azimuth angle A and is also the maximum slope for the pixel among all directions, which can be calculated directly in ArcGIS using DEM.

Combining (B3)–(B6), the calculation of S_p can be given

$$S_p = \arctan[\tan S \cos(\varphi - A)] \tag{B.7}$$

where S_p varies from $-S$ (downslope direction) to S (upslope direction), which means that A_p is negative in the downslope direction ($\pi/2 < \varphi < 3\pi/2$). $H_{(\varphi)}$ in the upslope direction is then calculated as follows:

$$H_{(\varphi)} = \frac{\pi}{2} - \arctan\left(\frac{\mathrm{El}_1}{D_1}\right) + \arctan[\tan S \cos(\varphi - A)]. \tag{B.8}$$

In the downslope direction

$$\tan\theta_2 = \frac{\mathrm{El}_2}{D_2} = \frac{\mathrm{El}_{t2} - \mathrm{El}_s}{D_2} \tag{B.9}$$

$$\theta_{\mathrm{sp2}} = \frac{\pi}{2} - \theta_2 - |S_p| = \frac{\pi}{2} - \arctan\left(\frac{\mathrm{El}_2}{D_2}\right) + S_p \tag{B.10}$$

where E_{t2} is the elevation of the surrounding terrain in the downslope direction. When $\mathrm{El}_{t2} < \mathrm{El}_s$, the maximum sky opening angle $\theta_{\mathrm{sp2}} > (\pi/2) + S_p$. Because the sky can only be seen above the horizontal line, θ_{sp2} should be calculated as follows:

$$\theta_{\mathrm{sp2}} = \begin{cases} \dfrac{\pi}{2} - \arctan\left(\dfrac{\mathrm{El}_2}{D_2}\right) + S_p, & \theta_{\mathrm{sp2}} \le \dfrac{\pi}{2} + S_p \\ \dfrac{\pi}{2} - |S_p|, & \theta_{\mathrm{sp2}} > \dfrac{\pi}{2} + S_p \end{cases} \tag{B.11}$$

Combining (B8) and (B11), $H_{(\varphi)}$ in the up- and down-slope directions is given by (ΔEl is the elevation differences of surrounding terrain and targes, and D is their plane distance)

$$H_{(\varphi)} = \begin{cases} \dfrac{\pi}{2} - \arctan\left(\dfrac{\Delta El}{D}\right) + \arctan[\tan S \cos(\varphi - A)], \\ \quad \theta_{\mathrm{sp}} \le \dfrac{\pi}{2} + S_p \\ \dfrac{\pi}{2} - |\arctan[\tan S \cos(\varphi - A)]|, \\ \quad \theta_{\mathrm{sp}} > \dfrac{\pi}{2} + S_p. \end{cases} \tag{B.12}$$

Appendix C
Calculations for SVF

Similar to the slope coordinate system, the calculation of SVF (V_{sp}) under the plane coordinate system can be expressed as follows:

$$V_{\mathrm{sp}} = \frac{1}{\pi}\int_0^{2\pi}\int_0^{H_{(\varphi')}} K(\theta, \varphi)\sin\theta\cos\theta d\theta d\varphi \tag{C.1}$$

where $H_{(\varphi')}$ is the maximum sky opening angle of specific azimuth angle φ in the plane coordinate system. Referred to (9), the SVF for any azimuth angle φ over complex terrain can be

$$V_{\mathrm{sp}-K} = \frac{1}{\pi}\int_0^{2\pi}\int_0^{H_{(\varphi')}} K(\varphi)\sin\theta \times [\cos\theta\cos S + \sin\theta\sin S\cos(\varphi - A)]d\theta d\varphi. \tag{C.2}$$

Equation (B2) can be further simplified as follows:

$$V_{\mathrm{sp}-k} = \frac{1}{2\pi}\int_0^{2\pi} K(\varphi)\cos S\sin^2 H_{(\varphi')} + \sin S\cos(\varphi - A)\times(H_{(\varphi')} - \sin H_{(\varphi')}\cos H_{(\varphi')})d\varphi \tag{C.3}$$

where $H_{(\varphi')}$ can also be transformed to the slope coordinate system as follows:

$$H(\varphi) = H(\varphi') + \frac{\pi}{2} - \arctan[-1/\tan S\cos(\varphi - A)]. \tag{C.4}$$

Then, combined with (8), the SVF defined in the slope coordinate system can be derived from

$$V_{s-\mathrm{sp}} = \frac{1}{2\pi}\int_0^{2\pi}\sin^2(H(\varphi') + \frac{\pi}{2} - \arctan\left[-1/\tan S\cos(\varphi - A)\right])d\varphi. \tag{C.5}$$

Based on the shadow-casting algorithm, the open-source software urban multiscale environmental predictor provided another scheme for SVF calculation. The core algorithm can be expressed as follows:

$$V_{s-\mathrm{sc}} = \sum_{n=1}^{N}\left(X_n\frac{1}{\pi}\sin\frac{\pi}{180}\sin\frac{\pi(2\theta_n - 1)}{2n}\frac{360}{k_n}\right) \tag{C.6}$$

where X_n is the nth shadow binary pixel, N is the total number of shadow images (i.e., 653), θ represents the altitude angle, and k_n is the azimuth angle.

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REFERENCES

[1] M. Chen, G. R. Willgoose, and P. M. Saco, "Investigating the impact of leaf area index temporal variability on soil moisture predictions using remote sensing vegetation data," *J. Hydrol.*, vol. 522, pp. 274–284, Mar. 2015.

[2] Q. Liu, L. Gu, R. E. Dickinson, Y. Tian, L. Zhou, and W. M. Post, "Assimilation of satellite reflectance data into a dynamical leaf model to infer seasonally varying leaf areas for climate and carbon models," *J. Geophys. Research: Atmos.*, vol. 113, no. D19, pp. 1–7, Oct. 2008.

[3] D. Slevin, S. F. B. Tett, and M. Williams, "Multi-site evaluation of the Jules land surface model using global and local data," *Geoscientific Model Develop.*, vol. 8, no. 2, pp. 295–316, Feb. 2015.

[4] R. Wang, J. M. Chen, Z. Liu, and A. Arain, "Evaluation of seasonal variations of remotely sensed leaf area index over five evergreen coniferous forests," *ISPRS J. Photogramm. Remote Sens.*, vol. 130, pp. 187–201, Aug. 2017.

[5] D. Baldocchi, J. Finnigan, K. Wilson, K. T. Paw U, and E. Falge, "On measuring net ecosystem carbon exchange over tall vegetation on complex terrain," *Boundary-Layer Meteorol.*, vol. 96, nos. 1–2, pp. 257–291, Aug. 2000.

[6] R. A. Duursma, J. D. Marshall, and A. P. Robinson, "Leaf area index inferred from solar beam transmission in mixed conifer forests on complex terrain," *Agricult. Forest Meteorol.*, vol. 118, nos. 3–4, pp. 221–236, Sep. 2003.

[7] N. Liu, A. C. Oishi, C. F. Miniat, and P. Bolstad, "An evaluation of ECOSTRESS products of a temperate montane humid forest in a complex terrain environment," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112662.

[8] G.-R. Yu, X.-F. Wen, X.-M. Sun, B. D. Tanner, X. Lee, and J.-Y. Chen, "Overview of ChinaFLUX and evaluation of its eddy covariance measurement," *Agricult. Forest Meteorol.*, vol. 137, nos. 3–4, pp. 125–137, Apr. 2006.

[9] Y. Liu et al., "Satellite-derived LAI products exhibit large discrepancies and can lead to substantial uncertainty in simulated carbon and water fluxes," *Remote Sens. Environ.*, vol. 206, pp. 174–188, Mar. 2018.

[10] G. Wu et al., "The increasing contribution of greening to the terrestrial evapotranspiration in China," *Ecological Model.*, vol. 477, Mar. 2023, Art. no. 110273.

[11] J. M. Chen and T. A. Black, "Defining leaf area index for non-flat leaves," *Plant, Cell Environ.*, vol. 15, no. 4, pp. 421–429, May 1992.

[12] J. M. Chen, P. M. Rich, S. T. Gower, J. M. Norman, and S. Plummer, "Leaf area index of boreal forests: Theory, techniques, and measurements," *J. Geophys. Res., Atmos.*, vol. 102, no. D24, pp. 29429–29443, Dec. 1997.

[13] A. Gonsamo and J. M. Chen, "Improved LAI algorithm implementation to MODIS data by incorporating background, topography, and foliage clumping information," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 2, pp. 1076–1088, Feb. 2014.

[14] E. M. Luisa, B. Frédéric, and W. Marie, "Slope correction for LAI estimation from gap fraction measurements," *Agricult. Forest Meteorol.*, vol. 148, no. 10, pp. 1553–1562, Sep. 2008.

[15] G. Yin, L. Ma, W. Zhao, Y. Zeng, B. Xu, and S. Wu, "Topographic correction for Landsat 8 OLI vegetation reflectances through path length correction: A comparison between explicit and implicit methods," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8477–8489, Dec. 2020.

[16] J. M. Chen and J. Cihlar, "Plant canopy gap-size analysis theory for improving optical measurements of leaf-area index," *Appl. Opt.*, vol. 34, no. 27, p. 6211, Sep. 1995.

[17] A. Gonsamo and P. Pellikka, "Methodology comparison for slope correction in canopy leaf area index estimation using hemispherical photography," *Forest Ecology Manage.*, vol. 256, no. 4, pp. 749–759, Aug. 2008.

[18] H. Jin, A. Li, W. Xu, Z. Xiao, J. Jiang, and H. Xue, "Evaluation of topographic effects on multiscale leaf area index estimation using remotely sensed observations from multiple sensors," *ISPRS J. Photogramm. Remote Sens.*, vol. 154, pp. 176–188, Aug. 2019.

[19] J. Wen et al., "Characterizing land surface anisotropic reflectance over rugged terrain: A review of concepts and recent developments," *Remote Sens.*, vol. 10, no. 3, p. 370, Feb. 2018.

[20] D. Gu and A. Gillespie, "Topographic normalization of Landsat TM images of forest based on subpixel sun–canopy–sensor geometry," *Remote Sens. Environ.*, vol. 64, no. 2, pp. 166–175, May 1998.

[21] Y. Gao and W. Zhang, "A simple empirical topographic correction method for ETM+ imagery," *Int. J. Remote Sens.*, vol. 30, no. 9, pp. 2259–2275, May 2009.

[22] P. M. Teillet, B. Guindon, and D. G. Goodenough, "On the slope-aspect correction of multispectral scanner data," *Can. J. Remote Sens.*, vol. 8, no. 2, pp. 84–106, 1982.

[23] G. Yin et al., "PLC: A simple and semi-physical topographic correction method for vegetation canopies based on path length correction," *Remote Sens. Environ.*, vol. 215, pp. 184–198, Sep. 2018.

[24] J. D. Shepherd and J. R. Dymond, "Correcting satellite imagery for the variance of reflectance and illumination with topography," *Int. J. Remote Sens.*, vol. 24, no. 17, pp. 3503–3514, Jan. 2003.

[25] Y. Chen, A. Hall, and K. N. Liou, "Application of three-dimensional solar radiative transfer to mountains," *J. Geophys. Res., Atmos.*, vol. 111, no. D21, pp. 1–13, Nov. 2006.

[26] R. Richter, T. Kellenberger, and H. Kaufmann, "Comparison of topographic correction methods," *Remote Sens.*, vol. 1, no. 3, pp. 184–196, Jul. 2009.

[27] S. A. Soenen, D. R. Peddle, and C. A. Coburn, "SCS+C: A modified sun-canopy-sensor topographic correction in forested terrain," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 9, pp. 2148–2159, Sep. 2005.

[28] J. A. Smith, T. L. Lin, and K. J. Ranson, "The Lambertian assumption and Landsat data," *Photogramm. Eng. Remote Sens.*, vol. 46, pp. 1183–1189, Sep. 1980.

[29] D. H. Reeder, "Topographic correction of satellite images: Theory and application," Dissertation/Thesis, Jul. 2002, vol. 1, no. 2, pp. 1–65.

[30] L. Blesius and F. Weirich, "The use of the minnaert correction for land-cover classification in mountainous terrain," *Int. J. Remote Sens.*, vol. 26, no. 17, pp. 3831–3851, Sep. 2005.

[31] A. A. Bayanuddin et al., "Comparison of topographic correction methods on SPOT-6/7 multispectral data in Indonesia: An initial evaluation in hilly to undulating area," in *Proc. 7th Geoinformation Sci. Symp.*, vol. 12082, Dec. 2021, pp. 1–12.

[32] N. Ghasemi, A. Mohammadzadeh, and M. R. Sahebi, "Assessment of different topographic correction methods in ALOS AVNIR-2 data over a forest area," *Int. J. Digit. Earth*, vol. 6, no. 5, pp. 504–520, Sep. 2013.

[33] J. Wu, M. E. Bauer, D. Wang, and S. M. Manson, "A comparison of illumination geometry-based methods for topographic correction of QuickBird images of an undulant area," *ISPRS J. Photogramm. Remote Sens.*, vol. 63, no. 2, pp. 223–236, Mar. 2008.

[34] M.-L. Gao, W.-J. Zhao, Z.-N. Gong, H.-L. Gong, Z. Chen, and X.-M. Tang, "Topographic correction of ZY-3 satellite images and its effects on estimation of shrub leaf biomass in mountainous areas," *Remote Sens.*, vol. 6, no. 4, pp. 2745–2764, Mar. 2014.

[35] S. Hantson and E. Chuvieco, "Evaluation of different topographic correction methods for Landsat imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 13, no. 5, pp. 691–700, Oct. 2011.

[36] J. Dozier and J. Frew, "Rapid calculation of terrain parameters for radiation modeling from digital elevation data," *IEEE Trans. Geosci. Remote Sens.*, vol. 28, no. 5, pp. 963–969, Sep. 1990.

[37] H. Yin, B. Tan, D. Frantz, and V. C. C. Radeloff, "Integrated topographic corrections improve forest mapping using Landsat imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 108, Apr. 2022, Art. no. 102716.

[38] A. Li, Q. Wang, J. Bian, and G. Lei, "An improved physics-based model for topographic correction of Landsat TM images," *Remote Sens.*, vol. 7, no. 5, pp. 6296–6319, May 2015.

[39] F. Li et al., "A physics-based atmospheric and BRDF correction for Landsat data over mountainous terrain," *Remote Sens. Environ.*, vol. 124, pp. 756–770, Sep. 2012.

[40] Z. Zhang, G. He, X. Zhang, T. Long, G. Wang, and M. Wang, "A coupled atmospheric and topographic correction algorithm for remotely sensed satellite imagery over mountainous terrain," *GIScience Remote Sens.*, vol. 55, no. 3, pp. 400–416, May 2018.

[41] W. Fan, J. M. Chen, W. Ju, and G. Zhu, "GOST: A geometric-optical model for sloping terrains," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 9, pp. 5469–5482, Sep. 2014.

[42] W. Fan, J. Li, and Q. Liu, "GOST2: The improvement of the canopy reflectance model GOST in separating the sunlit and shaded leaves," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 4, pp. 1423–1431, Apr. 2015.

[43] J. M. Chen and S. G. Leblanc, "A four-scale bidirectional reflectance model based on canopy architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 35, no. 5, pp. 1316–1337, Sep. 1997.

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[44] A. Huang et al., "Development of a clear-sky 3D sub-grid terrain solar radiative effect parameterization scheme based on the mountain radiation theory," *J. Geophys. Res., Atmos.*, vol. 127, no. 13, pp. 1–20, 2022.

[45] J. Qi et al., "LESS: LargE-scale remote sensing data and image simulation framework over heterogeneous 3D scenes," *Remote Sens. Environ.*, vol. 221, pp. 695–706, Feb. 2019.

[46] Z. Jiang, A. Huete, K. Didan, and T. Miura, "Development of a two-band enhanced vegetation index without a blue band," *Remote Sens. Environ.*, vol. 112, no. 10, pp. 3833–3845, Oct. 2008.

[47] R. B. Myneni, F. G. Hall, P. J. Sellers, and A. L. Marshak, "The interpretation of spectral vegetation indexes," *IEEE Trans. Geosci. Remote Sens.*, vol. 33, no. 2, pp. 481–486, Mar. 1995.

[48] M. D. Steven, "The sensitivity of the OSAVI vegetation index to observational parameters," *Remote Sens. Environ.*, vol. 63, no. 1, pp. 49–60, Jan. 1998.

[49] Y. Zeng et al., "Optical vegetation indices for monitoring terrestrial ecosystems globally," *Nature Rev. Earth Environ.*, vol. 3, no. 7, pp. 477–493, May 2022.

[50] A. Mousivand, W. Verhoef, M. Menenti, and B. Gorte, "Modeling top of atmosphere radiance over heterogeneous non-lambertian rugged terrain," *Remote Sens.*, vol. 7, no. 6, pp. 8019–8044, Jun. 2015.

[51] T. Wang, G. Yan, X. Mu, Z. Jiao, L. Chen, and Q. Chu, "Toward operational shortwave radiation modeling and retrieval over rugged terrain," *Remote Sens. Environ.*, vol. 205, pp. 419–433, Feb. 2018.

[52] S. Wu, J. Wen, D. You, H. Zhang, Q. Xiao, and Q. Liu, "Algorithms for calculating topographic parameters and their uncertainties in downward surface solar radiation (DSSR) estimation," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 8, pp. 1149–1153, Aug. 2018.

[53] S. Medeiros, S. Hagen, J. Weishampel, and J. Angelo, "Adjusting LiDAR-derived digital terrain models in coastal marshes based on estimated aboveground biomass density," *Remote Sens.*, vol. 7, no. 4, pp. 3507–3525, Mar. 2015.

[54] Z. Huang et al., "Functionally dissimilar neighbours increase tree water use efficiency through enhancement of leaf phosphorus concentration," *J. Ecology*, vol. 110, no. 9, pp. 2179–2189, Sep. 2022.

[55] X. Ni et al., "Decline in nutrient inputs from litterfall following forest plantation in subtropical China," *Forest Ecology Manage.*, vol. 496, Sep. 2021, Art. no. 119445.

[56] R. Siegel and J. R. Howell, *Thermal Radiation Heat Transfer*, 3rd ed. Bristol, PA, USA: Hemisphere Publishing, 1992.

[57] J. E. Hay, "Solar energy system design: The impact of mesoscale variations in solar radiation," *Atmosphere-Ocean*, vol. 21, no. 2, pp. 138–157, Jun. 1983.

[58] J. M. Chen, T. A. Black, D. T. Price, and R. E. Carter, "Model for calculating photosynthetic photon flux densities in forest openings on slopes," *J. Appl. Meteorol.*, vol. 32, no. 10, pp. 1656–1665, Oct. 1993.

[59] I. Sola, M. González-Audícana, J. Álvarez-Mozos, and J. L. Torres, "Synthetic images for evaluating topographic correction algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 3, pp. 1799–1810, Mar. 2014.

[60] C. J. Tucker, "Red and photographic infrared linear combinations for monitoring vegetation," *Remote Sens. Environ.*, vol. 8, no. 2, pp. 127–150, May 1979.

[61] K. Zakšek, K. Oštir, and Ž. Kokalj, "Sky-view factor as a relief visualization technique," *Remote Sens.*, vol. 3, no. 2, pp. 398–415, Feb. 2011.

[62] J. Wen, Q. Liu, Q. Liu, Q. Xiao, and X. Li, "Parametrized BRDF for atmospheric and topographic correction and albedo estimation in Jiangxi rugged terrain, China," *Int. J. Remote Sens.*, vol. 30, no. 11, pp. 2875–2896, Jun. 2009.

[63] W. Georg, H. Albin, N. Georg, S. Katharina, T. Enrico, and Z. Peng, "On the energy balance closure and net radiation in complex terrain," *Agricult. Forest Meteorol.*, vols. 226–227, pp. 37–49, Oct. 2016.

[64] H. Mannstein, "The interpretation of albedo measurements on a snow-covered slope," *Arch. Meteorol., Geophysics, Bioclimatology Ser. B*, vol. 36, no. 1, pp. 73–81, Mar. 1985.

[65] R. Wang, J. M. Chen, X. Luo, A. Black, and A. Arain, "Seasonality of leaf area index and photosynthetic capacity for better estimation of carbon and water fluxes in evergreen conifer forests," *Agricult. Forest Meteorol.*, vol. 279, Dec. 2019, Art. no. 107708.

[66] D. Haboudane, "Hyperspectral vegetation indices and novel algorithms for predicting green LAI of crop canopies: Modeling and validation in the context of precision agriculture," *Remote Sens. Environ.*, vol. 90, no. 3, pp. 337–352, Apr. 2004.

[67] D. R. A. Almeida et al., "Monitoring the structure of forest restoration plantations with a drone-LiDAR system," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 79, pp. 192–198, Jul. 2019.

[68] D. R. A. D. Almeida et al., "Monitoring restored tropical forest diversity and structure through UAV-borne hyperspectral and LiDAR fusion," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112582.

[69] H. Ma, J. Song, J. Wang, Z. Xiao, and Z. Fu, "Improvement of spatially continuous forest LAI retrieval by integration of discrete airborne LiDAR and remote sensing multi-angle optical data," *Agricult. Forest Meteorol.*, vols. 189–190, pp. 60–70, Jun. 2014.

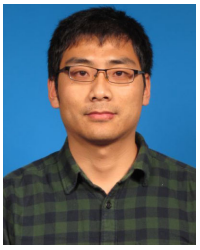
[70] Z. H. Jiao, H. Ren, X. Mu, J. Zhao, T. Wang, and J. Dong, "Evaluation of four sky view factor algorithms using digital surface and elevation model data," *Earth Space Sci.*, vol. 6, no. 2, pp. 222–237, 2019.

[71] J. Manners, S. B. Vosper, and N. Roberts, "Radiative transfer over resolved topographic features for high-resolution weather prediction," *Quart. J. Roy. Meteorological Soc.*, vol. 138, no. 664, pp. 720–733, Apr. 2012.

[72] F. Lindberg and C. S. B. Grimmond, "The influence of vegetation and building morphology on shadow patterns and mean radiant temperatures in urban areas: Model development and evaluation," *Theor. Appl. Climatol.*, vol. 105, nos. 3–4, pp. 311–323, Oct. 2011.

[73] H. Jin et al., "Intercomparison and validation of MODIS and GLASS leaf area index (LAI) products over mountain areas: A case study in southwestern China," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 55, pp. 52–67, Mar. 2017.

[74] M. Adam, M. Urbazaev, C. Dubois, and C. Schmullius, "Accuracy assessment of GEDI terrain elevation and canopy height estimates in European temperate forests: Influence of environmental and acquisition parameters," *Remote Sens.*, vol. 12, no. 23, p. 3948, Dec. 2020.



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